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Research Paper



# Microplastics and heavy metals emissions from healthcare waste management: A comparative life cycle assessment

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ABSTRACT

*Healthcare Waste Management* (HCWM) contributes to environmental pollution and health hazards. There are several established approaches to measure environmental and health impacts from HCWM. Amidst, *Life Cycle Assessment* (LCA) is one of the most popular tools to evaluate such impacts. This research was conducted to compare environmental impacts, and *Microplastics* (MPs) and *Heavy Metals* (HMs) emissions through LCA from HCWM between public and private *Healthcare Facilities* (HCFs) in Chattogram, Bangladesh. Five case studies were conducted based on the size of HCFs including one ideal scenario (developed with minimum impacts). The LCA was evaluated by SimaPro version 9.5.0.0 software and the ReCiPe 2016 method. Results revealed that private HCFs highly contributed to the eighteen studied impact categories. The main impact categories were human toxicity, global warming, ecotoxicity, eutrophication, acidification and resource scarcity. The existing common *Healthcare Waste* (HCW) disposal options were waste transport, open burning, recycling and dumping. The ranges of major impacts from HCWM were global warming (0.588–1.122 kg $CO_2$ eq), followed by human non-carcinogenic toxicity (0.002–0.003 kg $CO_2$ eq.) and ecotoxicity (0.000–0.001 kg $CO_2$ eq.). It demonstrated that dumping of HCW, landfill leachate, open burning/ incineration ash and recycling processes along with wastewater of recycling plants are the potential sources of HMs (up to 14480 mg/kg waste) and MPs (1906–7223 items/kg waste) pollution. This study demonstrates the potential levels of environmental impacts along with HMs and MPs emissions from the HCWM system, which will facilitate the relevant authorities and policy makers to adopt the required strategies.

## 1. Introduction

*Healthcare Waste* (HCW) is a broad topic that covers all sorts of solid wastes generated from *Healthcare Facilities* (HCFs) such as hospitals, clinics, diagnostic centers, labs, veterinary clinics, research centers and nursing homes (Fan et al., 2022; Hossain and Alam, 2012, Janik-Karpinska et al., 2023). HCW is also known as medical, hospital, clinical and biomedical waste. It contains infectious agents such as discarded cultures, stocks, specimens, tissue, blood, body fluids, sharps, Hepatitis B virus, Hepatitis C virus and HIV. This type of wastes is ranging from 10 % to 15 % of the total waste; therefore, HCWs are called dangerous/hazardous/biohazardous waste (Alam et al., 2015; Barua et al., 2015; Hossain and Alam, 2013). The percentage of infectious waste varied in HCW based on sources and regions. Mid December 2019, COVID-19 pandemic broke out in Wuhan, China and rapidly spread out throughout the world in 2020. This virus can transmit from person to person through direct physical contact, air and other potential ways (Belhadi et al., 2020; Kumar et al., 2021; Nabavi-Pelesaraei et al., 2022). Coronavirus is contagious and very dangerous, so people started to wear several *Personal Protective Equipment* (PPE) such as aprons, facemasks, goggles and hand gloves. Most of these PPEs, especially single use facemasks are globally used in large quantities and then discarded (Ajaj et al., 2023; Kokalj et al., 2022; Snigdha et al., 2023; Zhao et al., 2022). Such emerging wastes are responsible for booming the volume of total HCW. Therefore, it is a very challenging task to safely handle and dispose of such continuously growing waste piles (Friedericy et al.,

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2022; Kumar et al., 2021; Türkmen, 2022). Mishandling of HCW is responsible for spreading infectious agents (HIV/AIDS, COVID-19, Hepatitis B and C viruses) and unwanted accidents from sharp materials (Alam et al., 2014; Ghodrat et al., 2017; Hossain et al., 2013). Therefore, the HCW generation rate and its composition is very important for safe disposal.

Generally, HCW generation rate varies based on sources and regions. Previous studies demonstrated HCW generation rate 0.36–0.67 kg/cap/day and 1.3–3.0 kg/bed/day in developed countries while 2.17–2.63 kg/cap/day and 1.50–3.10 kg/bed/day in developing countries (Alam et al., 2013; Deepak et al., 2022; Hossain and Alam, 2012; Zhou et al., 2024). A major fraction of HCW is food waste in developing countries; however, it contains a significant amount of plastics & rubber, metals, glass, and small metallic tools. HCW consists of 6 %–20 % *Solid Plastic Waste* (SPW). Similarly, a large part of PPE is made of polymers. Therefore, the percentage of SPW greatly increased in HCW since COVID-19 out-break (Klemes et al., 2020; Kokalj et al., 2022; Liu et al., 2022; Rehman et al., 2025; Stapleton et al., 2023). These SPWs gradually degrade during recycling, treatment and disposal of HCW. Thus, macroplastics (<10 mm), *Microplastics* (MPs) (<5 mm) and nanoplastics (<1 μm) are generated from SPW throughout its life cycle (Alam et al., 2018; Ajaj et al., 2023; Shen et al., 2021; Zhao and You, 2022, 2023). With the passage of time, these MPs are distributed to urban environments and adjacent water bodies which subsequently reach to seas and oceans (Alam et al., 2019; Brown et al., 2023; Gallo et al., 2018; Liu et al., 2020; Silva et al., 2021). MPs are ingested by marine fish and then enter food chains. Finally, MPs are digested by human beings through food chains (Suzuki et al., 2022; Woods et al., 2021; Yang et al., 2021; Zhao and You, 2022; Zhong et al., 2024b).

Similarly, non-organic HCW wastes contain *Heavy Metals* (HMs) which are blended during their manufacture (Alam et al., 2018; Alam and Qiao, 2020; Alam et al., 2024b; Zahoor et al., 2023). The great challenge of HCW incineration is the emission of toxic HMs (e.g., Cr, Cd, Pb and Hg), dioxins, furans, and *Persistent Organic Pollutants* (POP). Incineration cannot destroy metallic components of HCW, so HMs are concentrated in the bottom ash (Adama et al., 2016; Alam et al., 2024a; Zhao et al., 2009; Zheng et al., 2024b). Besides, illegal dumping and landfilling release HMs to soil that subsequently reach adjacent water bodies and groundwater. Therefore, HCW disposal is a major environmental and health concern (Alam et al., 2014; Voudrias, 2023; Zheng et al., 2024a). Recently, management of landfill leachate and incineration ash has been under continuous scrutiny and control due to such concern (Adama et al., 2016; Liu and Terumasa, 2022; Zhong et al., 2024a).

Though COVID-19 wastes and general HCW are separately and safely handled and disposed in developed countries; however, it is often mixed up with *Municipal Solid Waste* (MSW) in developing countries (Ahmad et al., 2021; Alam and Qiao, 2020; Dey and Ray, 2023; Zhao et al., 2021). Therefore, it is a timely demand to develop a cost-effective and eco-friendly strategy for HCW disposal in developing countries focusing local scenarios. The application of *Life Cycle Assessment* (LCA) tool can support in this case for developing waste management strategy. LCA based studies were commenced in 1969 and developed in the 1980s. Recently, LCA is considered as the most popular tool to evaluate the environmental footprints and cost analysis of MSW management and disposal (Cavalloro et al., 2024; Dey and Ray, 2020, 2021; Maria et al., 2021; Muñoz et al., 2018; Piekarska et al., 2023; Sonibare et al., 2019; Song et al., 2017). However, it was first applied in HCW in 2006 by Qing et al. (2006). The application of LCA in HCW is still limited in developed countries like the U.S., the U.K., China, Brazil and E.U. countries (Nabavi-Pelesaraei et al., 2022; Soares et al., 2013; Thiel et al., 2015; Zhao et al., 2009) and very rare in developing countries (Alam and Mosharraf, 2019; Ali et al., 2016; Kumar et al., 2021).

Till now, several LCA studies were conducted to measure environmental impacts of *Healthcare Waste Management* (HCWM), and recently added facemasks disposal (Ajaj et al., 2023; Deepak et al., 2022; Türkmen, 2022; Zhao et al., 2021). Several LCA studies were performed comparing incineration and non-incineration treatment of HCW (Ahmad et al., 2019; Diaz et al., 2005; Ghodrat et al., 2017; Qing et al., 2006; Zhao et al., 2009), pyrolysis treatment (Li et al., 2022; Yousef et al., 2022), disposable medical facemasks (Kokalj et al., 2022; Türkmen, 2022; Yin et al., 2023), life cycle costing and economic assessment (Atia et al., 2020; Hong et al., 2018), chemical disinfection (Soares et al., 2013), recycling (Liu et al., 2022; Shan et al., 2023; Yin et al., 2023), reusable plastics/sharps, disposable materials and surgical instruments (Baruffaldi et al., 2019; Campion et al., 2015; Friedericy et al., 2022; Leiden et al., 2020; McPherson et al., 2019; Snigdha et al., 2023), laboratory/ pathological test (Gordon et al., 2021; Ji et al., 2022; McAlister et al., 2021,2022), wastewater treatment (Lutterbeck et al., 2020; Wang et al., 2020) and resources consumptions in HCFs (Twomey, 2011). What are the environmental emissions from HCWM based on the size of HCFs and operational functions along with focusing on HMs and MPs emissions during HCW disposal and treatment is still unexplored?

According to literature, there is no published research report focusing on LCA of cumulative disinfection, recycling, illegal disposal (open burning and dumping), treatment and sanitary landfilling of HCW (Alam and Mosharraf, 2019; Alam and Qiao, 2020; Chen et al., 2014). LCA study in HCWM is a challenging task due to the absence of a uniform management system and database based on local scenarios, especially in developing countries. Therefore, it is a timely demand to conduct a study on HCWM to explore the differences between public and private HCFs in terms of environmental footprints from HCW disposal system. This LCA study was conducted to explore the HCWM system with minimum environmental impacts including potential MPs and HMs emissions.

The findings of this research will facilitate in identifying HCF (specifically based on public and private along with size) with the least environmental footprints from waste disposal options. Besides, the HMs and MPs emissions scenarios from different HCFs will support the management authorities and relevant policy makers to adopt the required strategies so that environmental emissions can be minimized.

## 2. Methodology

### 2.1. Background information

#### 2.1.1. Primary data collection

The entire *Chattogram City Corporation* (CCC), Bangladesh was considered as a study area. It is situated within 22°–14′ and 22°–24′–30″ N Latitude and between 91°–46′ and 91°–53′ E Longitude. The city is located on the North bank of the river Karnafully. Two categories of respondents were selected with a view to collecting data such as HCW generator i.e. HCFs and HCW management authorities i.e. CCC (public) and *Innovation Sheva Sangstha* (ISS) (private). HCW generator respondents were selected based on their importance and size, and the quantity of HCW they generate. The existing HCFs of CCC were classified into four categories based on their size and operating functions including (i) large scale hospital, (ii) medium scale clinic, (iii) small scale clinic/ diagnostic center, and (iv) rural clinic/ diagnostic center. The data were collected from selected HCFs namely *Chattogram Medical College and Hospital* (CMCH), *University of Science and Technology, Chattogram* (USTC), *Memon Maternity Hospital* (MMH), *National Hospital* (NH), *City Corporation Charitable Clinic* (CCCC), *Metro Diagnostic Center* (MDC), *Patiya Upazila Health Complex* (PUHC), and *Chowdhury Hat Nursing Home* (CHNH) because they can represent all the available HCFs in Chattogram. There are many differences between public and private HCFs in terms of operating functions and providing treatment services. Public HCFs have more functions and provide services to more patients than private one.

Two sets of semi-structured questionnaires were designed – one for the HCW generator and the other for the HCW management authorities were designed based on gathered experiences from empirical field visit

and literature review. Questionnaire survey was conducted by fixing a prior appointment schedule with selected HCFs. Then survey was conducted physically and through face-to-face interview. During field data collection, the questionnaires were reformed based on respondents' feedback (when it was required). In total, 8 HCFs were surveyed from which wastes are either collected by CCC or ISS for collecting data for this investigation. Besides, the personnel of CCC and ISS were interviewed to assess their opinions. Several times field visits were conducted with a view to perform questionnaire surveys and waste measurement over the study period. During the field visit, both HCF authorities and HCW management authority were interviewed. In addition, a qualitative survey was conducted among the doctors, nurses and workers to make an informative presentation of the quantitative survey findings.

The numeric data of HCW generation was calculated by measuring weight of the daily generated total HCW and individual composition of waste in dry season. At first, the total weight of the HCW from each HCF was measured by an electric balance and then recorded. The whole HCW was separated according to their physical compositions including plastics, rubbers & packaging, papers & packaging, clothes & textiles, steel/ metals, glass & ceramics, drugs/ medicines, chemicals/ radioactive, body fluids/ blood, body parts, cardboard/ woods/ leaves, and food waste. Then each separated composition of the HCW was measured and recorded. This process was followed in three consecutive days. Finally, the percentage of each composition was calculated as dividing whole HCW weight by individual composition weight and then multiplying by 100 (see details in Table 1). Besides, from each selected HCF, the daily generated plastic wastes were separated and measured by weight using electric balance and recorded. In addition, the activities of CCC and ISS related to HCW handling and management were deeply observed at the time of field visit. The findings are discussed in Section 2.1.1 (Supporting Information), and Table SI-1, SI-2, SI-3 and SI-4 (Supporting Information).

#### 2.1.2. Healthcare waste management system

HCW solely contributes 2 %–3% to the daily generated MSWs volume in Chattogram City, Bangladesh. However, there is no similarity in HCW generation, handling, management and disposal of different categories of HCFs (public and private). Furthermore, there is absence of proper record keeping of resources consumption and daily HCW generation. The details about the existing HCWM system are shown in Fig. SI-1 (Supporting information). Bangladesh government has formulated several rules for medical waste, hazardous materials, e-waste, solid waste and environment conservation such as (i) Medical Waste (Management and Processing) Rules, 2008, (ii) Solid Waste Management Rules 2021, and (iii) Environment Conservation Rules 2023 under the Bangladesh Environmental Protection Act, 1995. Following these regulations, public HCFs are practicing color coding bins but not properly followed in private HCFs. Therefore, most of the HCFs mix up the daily generated HCW that convert non-infectious waste into infectious. From sources, about 30 % HCW is collected by private organizations, while 35 % by Community Based Organizations (CBOs) and 25 % by HCFs directly discharge to municipal dustbins. Besides, about 10 %–15 % HCW are illegally disposed of on roads, drains and fallow lands (Alam et al., 2014, 2015; Hossain and Alam, 2012; Hossain et al., 2013). Reusable metallic instruments are reused through steam/microwave sterilization and chemical disinfection. HCW are manually handled and segregated at both sources and dumpsite without wearing any protective gears. In addition, hardly sharp and infectious HCW are disposed of by bucket type incinerators without energy recovery. From the HCFs about 40 %–55 % recyclable HCW such as plastics, papers, metals and glass directly go to scrap shops. Similarly, about 20 %–25 % HCW are disposed of through open burning while 10 %–20 % are buried (Alam and Hossain, 2013; Alam and Hossain, 2015; Hossain and Alam, 2012). The rest HCWs are transported to the dumpsite and disposed of through open burning and crude dumping. Again, the remaining recyclable HCW are picked up by scavengers from the dumpsite and sold to recycling shops. However, the HCWM services of both municipal and private organizations are not at satisfactory level due to the absence of energy recovery, leachate collection and gaseous emissions control system at disposal site. Therefore, the existing HCWM is polluting surrounding air, water and soil environments. However, the demand for private initiative services and adopting modern HCW treatment technologies are increasing because the municipality is not capable to handle HCW separately and control continuously increasing environmental pollution.

### 2.2. Life cycle assessment

The environmental and health impacts from HCW management and their treatment and disposal were evaluated using LCA according to the guidelines of "ISO 14040: Principles and Framework" and "ISO 14044: Requirements and Guidelines" (Ji et al., 2022; Leiden et al., 2020; Yin et al., 2023). This study adopted four steps: (i) goal and scope definition; (ii) development of system boundary, (iii) inventory data collection, (iv) impact assessment, (v) interpretation.

#### 2.2.1. Goal and scope definition

This study aimed to compare the HCWM system between public and private HCFs based on the above mentioned four categories along with the proposed ideal one (with minimum environmental impacts), in terms of environmental footprints, emphasizing MPs and HMs pollution. This research covered all types of solid HCW transport, disinfection, recycling, open burning, incineration, crude dumping, burial, composting and sanitary landfill. However, the management of wastewater from HCFs and life cycle cost are excluded because these data are not available in their records.

HCW is usually a non-homogenous mixture of solid wastes. The

**Table 1**
Generated healthcare wastes categorization from the studied healthcare facilities.

| HCW type | CMCH (%) | USTC (%) | MMH (%) | NH (%) | CCCC (%) | MDC (%) | PUHC (%) | CHNH (%) |
|---|---|---|---|---|---|---|---|---|
| Plastics, rubbers & packaging | 7.89 | 14.32 | 13.35 | 12.07 | 18.36 | 18.24 | 12.43 | 13.29 |
| Papers & packaging | 13.56 | 14.65 | 14.78 | 14.53 | 18.40 | 17.61 | 14.24 | 14.76 |
| Cloths & textiles | 9.68 | 9.77 | 9.30 | 9.04 | 13.32 | 13.27 | 9.88 | 10.16 |
| Steel/ metals | 6.05 | 6.36 | 5.70 | 6.20 | 9.42 | 8.98 | 6.89 | 6.50 |
| Glass & ceramics | 3.01 | 3.02 | 2.43 | 2.68 | 3.68 | 4.14 | 1.95 | 2.68 |
| Drugs/ medicines | 1.18 | 1.09 | 0.83 | 0.90 | 1.72 | 1.98 | 0.90 | 1.13 |
| Chemicals/ radioactive | 0.28 | 0.27 | 0.12 | 0.12 | 0.12 | 0.12 | 0.12 | 0.16 |
| Body fluids/ blood | 1.28 | 0.71 | 0.65 | 0.65 | 0.44 | 0.65 | 0.62 | 0.66 |
| Body parts | 1.42 | 1.01 | 0.44 | 0.80 | 0.80 | 0.00 | 0.80 | 0.45 |
| Cardboard/ woods/ leaves | 6.22 | 5.87 | 4.70 | 4.70 | 4.92 | 3.99 | 4.70 | 4.58 |
| Food waste | 49.44 | 42.93 | 47.7 | 48.31 | 28.82 | 31.01 | 47.47 | 45.68 |
| **Total waste (kg/day)** | **3285** | **580** | **280** | **390** | **25** | **165** | **130** | **38** |

**Note**: CMCH = Chittagong Medical College Hospital, USTC = University of Science and Technology, Chattogram, MMH = Memom Maternity Hospital, NH = National Hospital, CCCC = City Corporation Charitable Clinic, MDC = Metro Diagnostic Center, PUHC = Patiya Upazila Health Complex, and CHNH = Chowdhury Hat Nursing Home.

characteristics of HCW in the studied area was examined by our previous studies (Alam and Hossain, 2013; Alam and Qiao, 2020; Hossain and Alam, 2012; Hossain and Alam, 2013). However, a specific type of solid HCW with an average composition of the entire wastes generated daily in the selected HCFs is shown in Table 1, excluding wastewater and gaseous emissions. The assessment of these inputs will provide a comparison for establishing a sustainable HCWM system in HCFs. However, the functional unit of this study is disposal of one (1) kg HCW.

### 2.2.2. *Development of system boundary*

This study used a descriptive LCA framework to represent the HCWM system of Chattogram City including four case studies focusing the available management system based on their categories and one proposed system (see Fig. 1). These case studies represent the different disposal and treatment options of the entire HCWM generated from the selected HCFs. They also covered the HCW generation stage to final disposal point. Here, HCW is mostly segregated manually without wearing any protective gears that is a potential source of health hazards of the waste handlers. Besides, open burning and dumping with MSW are two commonly practiced HCW disposal options. The readily recyclable wastes (plastics, metals and glass) are informally recycled without pretreatment that is silently killing the health of the associated workers. In addition, the percentage of different categories of HCW disposal through open burning, dumping, composting, recycling, treatment and landfilling are included in these case studies and the developed ideal scenario. Fig. 1 (CMCH) shows the percentage of total HCW disposed of by different disposal options. Notably, the same amount of disinfected HCW is reused or recycled (ideal scenario), so it showed repeated percentage. Then the actual quantity of each physical composition disposed of by each disposal option is displayed. Similarly, the other HCFs scenarios are presented.

Here, the ideal HCWM system was proposed considering waste disposal options with minimum environmental impacts. The ideal system demonstrated comparatively lower environmental emissions than the existing HCWM systems. From Table 1, the numerical data of each physical composition was further calculated based on disposal options.

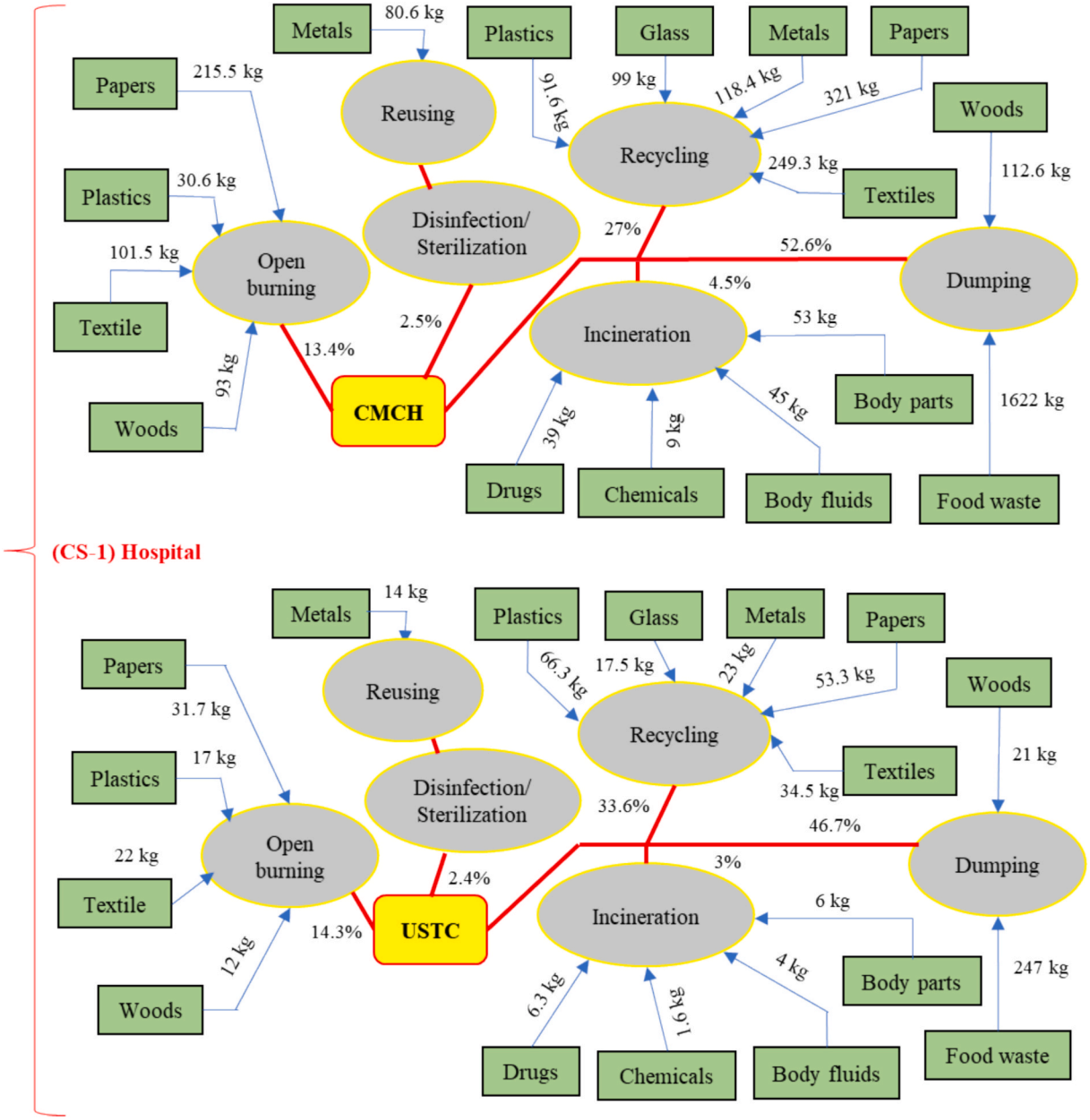



**Fig. 1.** Four case studies and one idea scenario adopted for life cycle assessment analysis in this study.

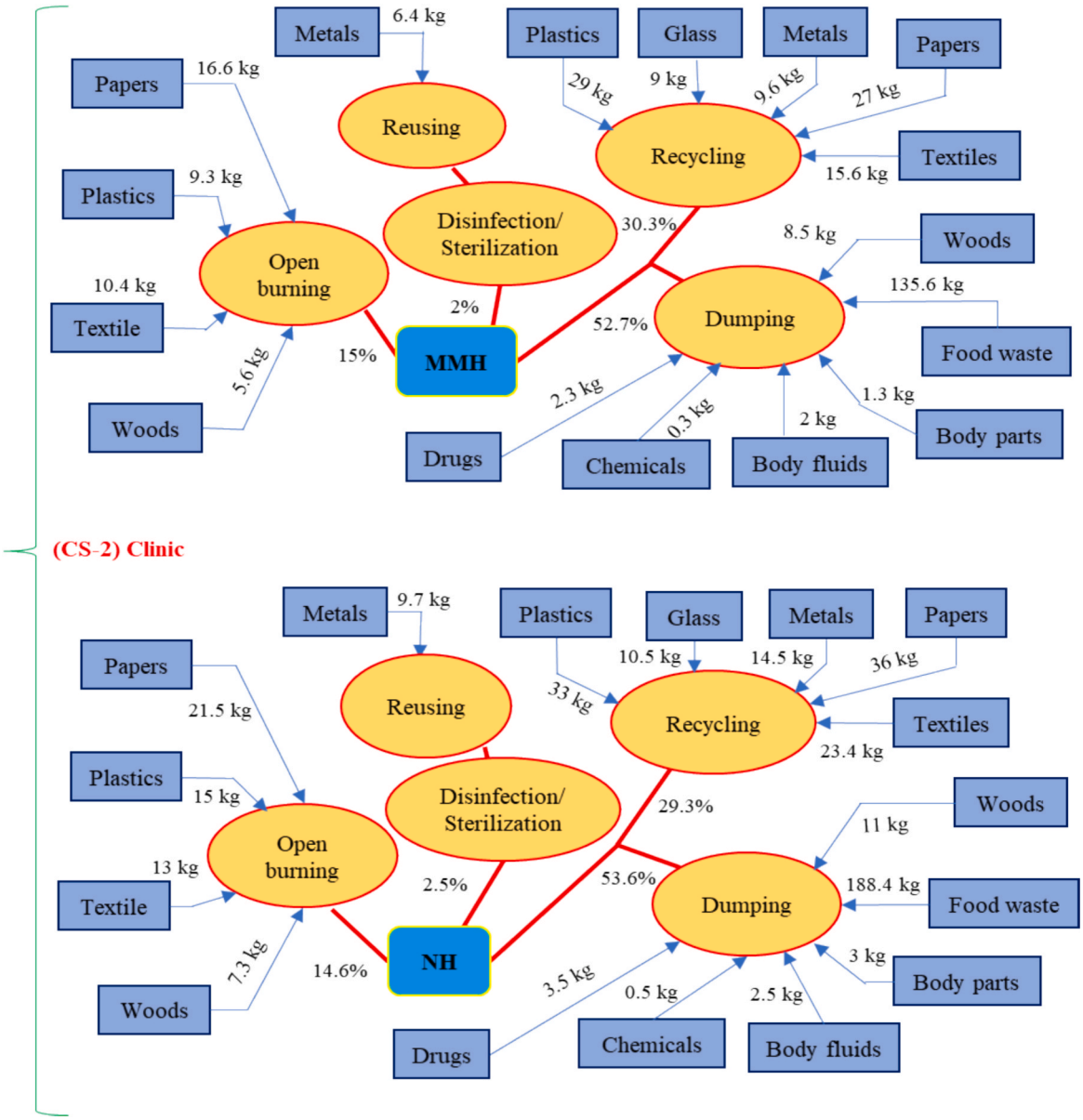



**Fig. 1.** (*continued*).

Here, several compositions were disposed of by more than one disposal option like plastic waste. Thus, the calculated percentage or numeric value of each physical composition by different disposal options was used in Fig. 1.

System boundary of any study depends on the goal and objectives of the study. The system boundary of this study is cradle to grave process. This study is confined within the activities of HCW management, treatment and disposal in HCFs. The impact of wastewater and its treatment is not considered here due to the unavailability of related data. The input in HCWM included solid waste, water, electricity, diesel and chemicals. The key output from the HCWM is the emissions to air, soil, wastewater/ leachate and solid waste (see Fig. 2). Open firing of HCW is arranged both in HCFs on-campus and dumping station for reducing wastes volume, that is liable for GHGs, PMs, VOCs, CO, NOx, $SO_2$ and dioxin/furans emissions to the atmosphere. The illegal dumping and landfilling of HCW contributes to soil (HMs) and water (toxic metals, N, P, COD) pollution. Similarly, the openly burned or incinerated HCWs generate HMs containing incineration ash. The external energy (electricity) and water are used in recycling of recyclable waste and steam sterilization for metallic reusable instruments, while bleaching powder in chemical disinfection. In addition, the diesel is used in incineration of HCWs. The incinerated ash is only dumped and there is no energy recovery system during incineration. In addition, there is no leachate and landfill gas collection system in the dumpsite and landfill.

### 2.2.3. *Life cycle inventory*

The steps followed for *Life Cycle Inventory* (LCI) in this study are guided by ISO standard 14040/44. The LCI data used in this study were collected from a combination of sources to ensure comprehensive and accurate representation of the HCWM system. The LCI is derived from the literature, government reports, expert opinions, physical interviews with responsible personnel of HCFs and the ecoinvent database. A detailed LCI data table is provided below to ensure transparency and reproducibility (see Table 2). Inventories generated through this work are expected to support similar future studies in other developing countries like India, Pakistan and Africa. The LCI of this study includes all the programs of HCWM that consume energy, water and materials from waste generation to final disposal stage. All the required data used in this study was collected from the HCFs and HCWM organizations. The used data covers the HCW collection, transportation, segregation,

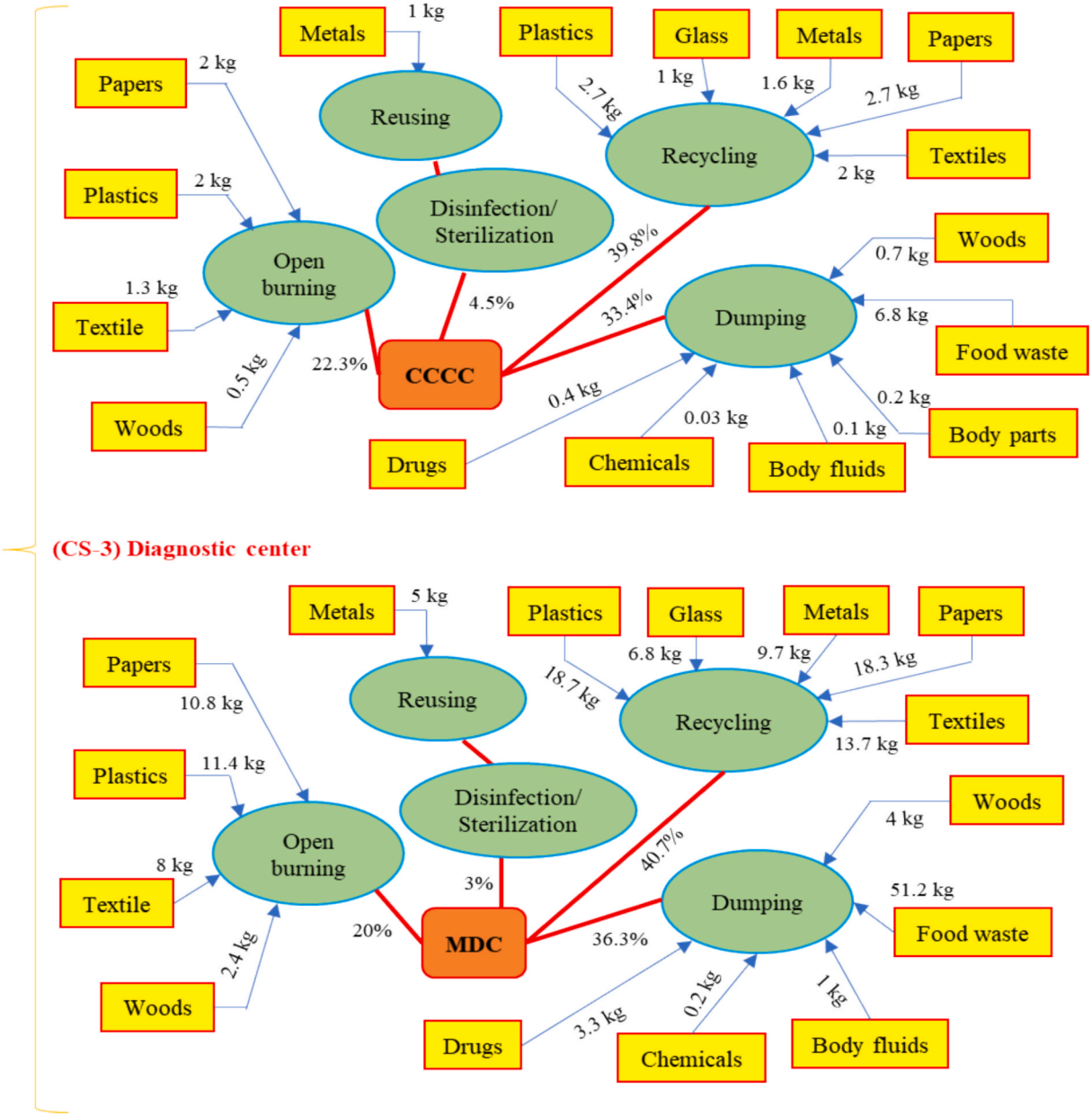



**Fig. 1.** (*continued*).

disinfection, illegal disposal, recycling, open burning and final disposal. However, the upstream processes are excluded in this study because the study focused on the HCWM system from the point of waste generation to final disposal. Upstream processes, such as the manufacturing of incinerators, containers, disinfectants, etc., were excluded to maintain a clear and manageable system boundary. Including these processes would significantly expand the scope of the study and introduce complexities related to the life cycle of these products. Besides, this study aimed to provide a focused analysis of the HCWM system within the defined boundaries. While upstream processes are important, they were considered outside the primary scope to ensure the study remained tractable and relevant to the specific context of HCWM.

The LCI includes HCW disposal but excludes upstream burdens associated with HCWM. The foreground data of raw materials and energy input were collected from our previous studies. The HCW was transported about 10–15 km from source to disposal point Table SI-5(a-f) (Supporting Information). Solid waste, energy, diesel, chemicals and water are the primary input for the HCWM system. The background data were from ecoinvent database in the environmental footprint platform. The MPs emissions from HCW disposal and treatment data were collected from direct field monitoring and compared with MPs emissions from same the amount of plastic wastes disposal. The MPs emissions from 1 kg of plastics wastes was measured by comparing with previous studies (Brown et al., 2023; Matee and Manyele, 2015; Nordahl et al., 2023; Shen et al., 2021; Silva et al., 2021; Stapleton et al., 2023; Yang et al., 2021; Zhang et al., 2019). Here, the plastics data were considered as input data for LCA software, and the MPs emissions (impact) counted as output data. It is very important to develop an LCA database for MPs emissions, however, this study used most of the MP's data from related literature. Therefore, it is currently not possible to develop a specialized MPs database. However, this study has made some recommendations in the conclusion section as future research direction. The inputs and outputs are expressed for each functional unit – 1 kg of waste. The inventory table is generated based on the aggregation of all inputs and output data collected throughout the life cycle stages of 1 kg of HCW. There are several types of treatment processes in HCWM which ease to

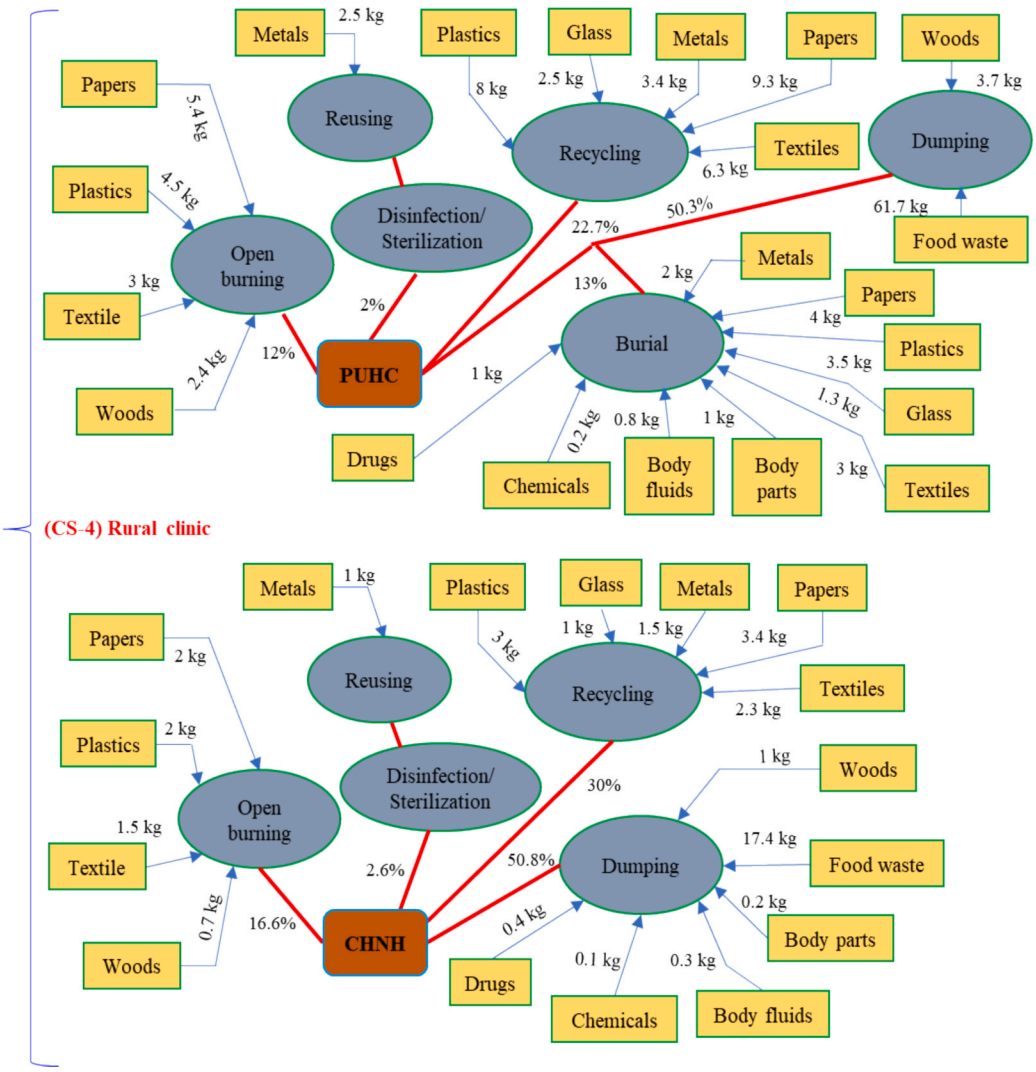



**Fig. 1.** (*continued*).

different compartments in the environment like air, water and soil.

### 2.2.4. Life cycle impact assessment

According to ISO 14040, the *Life Cycle Impact Assessment* (LCIA) tries to comprehend and assess the possible environmental impacts of HCWM systems, and their potential scale and relevance. The life cycle impacts at the midpoint and endpoint levels of this study were evaluated by SimaPro version 9.5.0.0 software. The ReCiPe 2016 Midpoint (H) V1.07/ World (2010) H and ReCiPe 2016 Endpoint (H) V1.07/ World (2010) H/A impact assessment methods were applied, respectively to calculate midpoint and endpoint environmental impacts connected to environmental emissions. The most common LCA software's include Simapro, Gabi and OpenLCA. Amidst, Simapro software was used in this study because it stands out among LCA software due to its comprehensive databases (e.g., Ecoinvent), flexibility for modeling complex systems, support for multiple impact assessment methods (like ReCiPe and TRACI), and compliance with ISO standards. It offers transparent calculations, user-friendly interfaces, and robust scenario analysis capabilities. Widely used in academia and industry, it ensures credibility, integrates with other tools, and provides strong documentation and support, making it ideal for reliable and detailed LCA studies. Besides, the ReCiPe 2016 method is superior because it provides updated, science-based characterization factors for both midpoint and endpoint impacts, offering more accurate and comprehensive assessments. It covers a wide range of environmental impacts and includes three perspectives (individualist, hierarchist, and egalitarian) for flexibility in decision-making. ReCiPe 2016 is widely recognized, aligns with international LCA standards, and incorporates improvements in modeling compared to older versions, making it ideal for modern sustainability analyses.

The best location for midpoint indications on an impact pathway is at the point where the mechanism remains the same for all flows classified under that effect category. In this case, characterization is the process of converting the LCI result to common units and combining the adapted results into an impact category. There are three impact assessment categories (human health, ecosystem and resources) at endpoint level and eighteen (18) impact categories at midpoint level. The midpoint 18 impact categories: Global Warming (GW), Stratospheric Ozone Depletion (SOD), Ionizing Radiation (IR), Ozone Formation, Human Health (OF, HH), Fine Particulate Matter Formation (FPMF), Ozone Formation, Terrestrial Ecosystems (OF, TE), Terrestrial Acidification (TA), Freshwater Eutrophication (Feu), Marine Eutrophication (Meu), Terrestrial Ecotoxicity (TE), Freshwater Ecotoxicity (Fec), Marine Ecotoxicity (Mec), Human Carcinogenic Toxicity (HCT), Human Non-carcinogenic Toxicity (HNCT), Land Use (LU), Mineral Resource Scarcity (MRS), Fossil Resource Scarcity (FRS), and Water Consumption (WC).

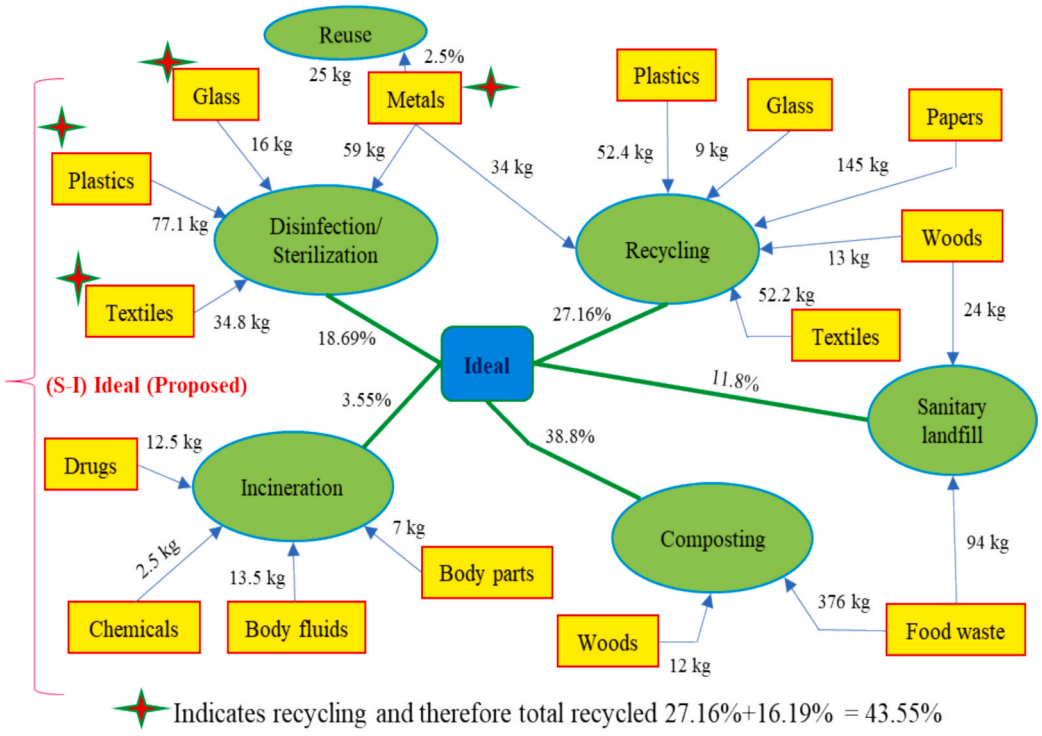



**Fig. 1.** (*continued*).

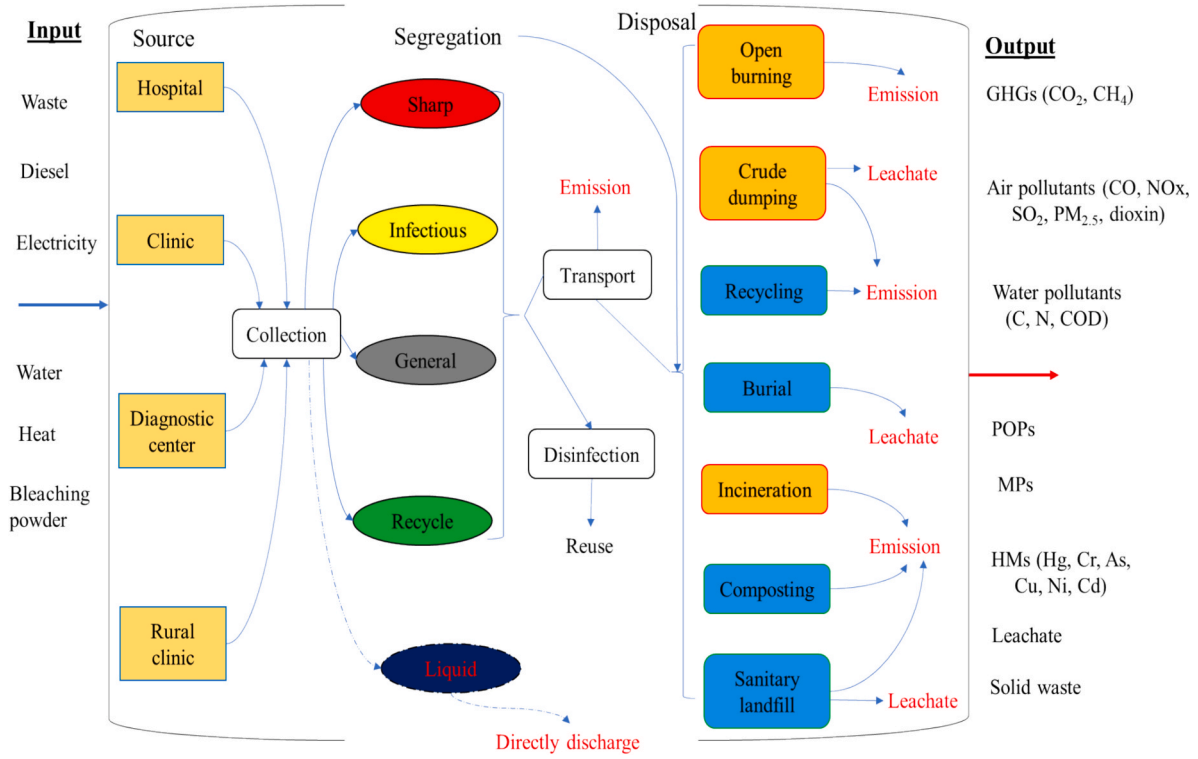



**Fig. 2.** System boundary for LCA study where red color indicate potential pollution and dotted line indicate not included (**Note**: POPs = Persistent organic pollutants, MP = Microplastics). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

### *2.2.5. Interpretation*

Sensitivity and contribution analyses were performed to interpret the LCIA results of this study. A sensitivity analysis was conducted to justify which environmental and health impact categories were sensitive to the change of the input parameters. In this study, sensitivity analysis of energy demand with varying transport (20 % increase and decrease) distance was conducted. It helps to identify the most interesting parameter that needs to be investigated in the LCA study.

Contribution analysis was performed to understand which input parameter of a specific HCW disposal and treatment process contributed

**Table 2**
Life cycle inventory data used in this study.

| Process/Activity | | Input/Output | Unit | Value | Source | Assumptions |
|---|---|---|---|---|---|---|
| Waste generation | Plastics | kg | 0.08 | Literature | Alam and Hossain, 2013; Hossain and Alam, 2012 | ±10 % |
| | Papers | kg | 0.15 | Literature | Alam and Hossain, 2013; Hossain and Alam, 2012 | ±10 % |
| | Textiles | kg | 0.10 | Literature | Alam and Hossain, 2013; Alam et al., 2015 | ±10 % |
| | Metals | kg | 0.07 | Literature | Alam and Hossain, 2013; Hossain and Alam, 2013 | ±10 % |
| | Glass | kg | 0.03 | Literature | Alam et al., 2015; Hossain and Alam, 2012 | ±10 % |
| | Food waste | kg | 0.50 | Literature | Alam and Hossain, 2013; Hossain and Alam, 2013 | ±10 % |
| Waste transport | Diesel | L | 0.05 | Government reports | Average fuel consumption for waste transport vehicles | ±15 % |
| Disinfection | Electricity | kWh | 0.10 | Ecoinvent database | Average energy consumption for disinfection processes | ±10 % |
| Recycling | Electricity | kWh | 0.08 | Ecoinvent database | Average energy consumption for recycling processes | ±10 % |
| Open burning | Emissions ($CO_2$) | kg | 0.50 | Expert opinions | Average emission factor for open burning | ±20 % |
| | Emissions (PM2.5) | kg | 0.02 | Expert opinions | Average emission factor for open burning | ±20 % |
| Incineration | Emissions ($CO_2$) | kg | 0.60 | Ecoinvent database | Average emission factor for incineration | ±15 % |
| | Emissions (HMs) | kg | 0.005 | Ecoinvent database | Average emission factor for incineration | ±15 % |
| Landfilling | Leachate | L | 0.03 | Government reports | Average leachate production for landfilling | ±10 % |
| | Emissions ($CH_4$) | kg | 0.01 | Ecoinvent database | Average emission factor for landfilling | ±10 % |

**Note:** The assumptions values in this table, such as ±10 % or ±15 %, typically represent the uncertainty or variability associated with the given values for inputs/outputs or emissions. The common methods adopted to determine these assumption values include literature review, government reports, ecoinvent database and expert opinions. Here, multiple studies and reports related to the specific waste management processes (e.g., waste generation, transport, recycling, etc.) were reviewed. If different studies report slightly different values for plastics waste generation, the ±10 % accounts for this range of variation. Similarly, government reports often provide standardized data on waste management processes, such as average fuel consumption for waste transport vehicles. Government data is usually based on large datasets and includes some inherent variability, which is captured by the ±15 % range. Besides, the Ecoinvent database provides detailed LCI data for various industrial processes, including energy consumption and emissions. Ecoinvent data is aggregated from multiple sources, and the ±10 % accounts for differences in process efficiencies, equipment types, or operational conditions. In addition, experts in the field provide estimates based on their experience and knowledge of specific processes, especially when empirical data is limited. Expert opinions are valuable when data is scarce, but they inherently include a higher degree of uncertainty, which is reflected in the ±20 % range.

most to the environmental pollution and health hazards.

## 3. Results and discussion

### 3.1. Comparative environmental impacts from the life cycle perspective

The inventory table was generated based on the aggregation of all inputs and output data collected throughout the life cycle stages of 1 kg of HCW. There are several types of disposal and treatment processes in HCWM which ease to different compartments in the environment like air, water and soil. Significant substances are listed with amount in Table SI-6(a-b) (Supporting Information).

This comparative study investigated 18 environmental impact categories in the HCWM system of the selected eight HCFs both public and private. The details about the environmental impacts from the HCWM are shown in Fig. 3 and Table SI-7 (Supporting Information). Results demonstrated that six impact categories were higher from USTC than CMCH in CS-1 where TE was the highest. In CMCH, the highest impact was HNCT (5.58 kg 1,4–dichlorobenzene (DCB)), followed by GW (0.73 kg $CO_2$ eq), Mec (0.32 kg 1,4-DCB) and Fec (0.24 kg 1,4-DCB). Here, SPW and waste papers mainly contributed to HNCT that come from open burning and recycling. Similarly, the highest impact was HNT (4.89 kg 1,4-DCB) from USTC followed by GW (0.70 kg $CO_2$ eq), Mec (0.28 kg 1,4-DCB), Fec (0.21 kg 1,4-DCB) and TE (0.13 kg 1,4-DCB). The SPW shared primarily to HNCT followed by wastepaper and textile wastes. Most of the studied environmental impacts were higher from CMCH due to it being the prime healthcare center in Chattogram and containing varied HCW compositions.

As medium scale HCF, MMH (public) showed comparatively higher environmental impacts than NH that might be for generating more food waste and others from MMH. The highest environmental impact from MMH was HNCT (5.44 kg 1,4-DCB), followed by GW (0.77 kg $CO_2$ eq), Mec (0.31 kg 1,4-DCB), Fec (0.24 kg 1,4-DCB) and TE (0.21 kg 1,4-DCB). The disposal of SPW and wastepaper mainly contributed to HNCT. Similarly, the highest impact was HNCT (5.51 kg 1,4-DCB) from NH, followed by GW (0.79 kg $CO_2$ eq), Mec (0.32 kg 1,4-DCB) Fec (0.24 kg 1,4-DCB) and TE (0.17 kg 1,4-DCB). Illegal disposal (open burning) and recycling of SPW and wastepaper is largely shared to HNCT in private NH. All the studied impact categories were higher in public clinics indicating the HCWM system of MMH is more responsible for environmental footprints.

Like CMCH and USTC, MDC demonstrated higher environmental impacts in most of the categories than CCCC (public clinic); however, CCCC contributed to higher impacts in case of IR, TE, LU and FRS (see Fig. 3). However, the highest impact was HNCT (3.29 kg 1,4-DCB), followed by GW (0.69 kg $CO_2$ eq), TE (1.79 kg 1,4-DCB), Mec (0.19 kg 1,4-DCB) and Fec (0.14 kg 1,4-DCB) from HCW disposal in CCCC. Here, HNCT was mainly shared by SPW, followed by waste transport, wastepaper and textile waste. In MDC, the highest impact was HNCT (3.64 kg 1,4-DCB), followed by GW (0.64 kg $CO_2$ eq), TE (0.36 kg 1,4-DCB), Mec (0.21 kg 1,4-DCB) and Fec (0.16 kg 1,4-DCB). Here, the HNCT impact was mainly shared by SPW from MDC, followed by wastepaper and textile waste. Most of the considered environmental impacts were higher from MDC in terms of small scale HCF. It was for generating relatively large quantities of HCW with a higher percentage of hazardous waste from MDC.

As rural HCF, CHNH (private clinic) showed higher environmental impacts than PUHC where eutrophication, ecotoxicity and human toxicity were almost similar (see Fig. 3). However, the highest environmental impact was HNCT (5.52 kg 1,4-DCB) from HCW disposal in PUHC, followed by GW (0.78 kg $CO_2$ eq), TE (0.38 kg 1,4-DCB), Mec (0.32 kg 1,4-DCB) and (0.24 kg 1,4-DCB). Here, transport of waste, and open burning and recycling of SPW is largely shared to HNCT. Similarly, the highest impact from CHNH was HNCT (5.27 kg 1,4-DCB), followed by TE (1.18 kg 1,4-DCB), GW (0.84 kg $CO_2$ eq), Mec (0.30 kg 1,4-DCB) and Fec (0.23 kg 1,4-DCB). HNCT impact was mainly contributed by SPW, followed by waste transport, wastepaper and textile waste in CHNH. In rural clinics, all the studied impact categories were higher in private clinics (CHNH), which means the existing HCWM system is liable for environmental pollution.

From the studied 18 impact categories, 15 impact categories were found comparatively higher from rural private HCF. It might be due to disposing of almost 100 % of HCW in an illegal and uncontrolled way. As the disposal of HCW is uncontrolled, therefore private HCFs had greater impact on several environmental factors such as climate change, ozone depletion, eutrophication, acidification, resource depletion, ecotoxicity and human toxicity.

The ideal (proposed) HCWM system demonstrated the lowest

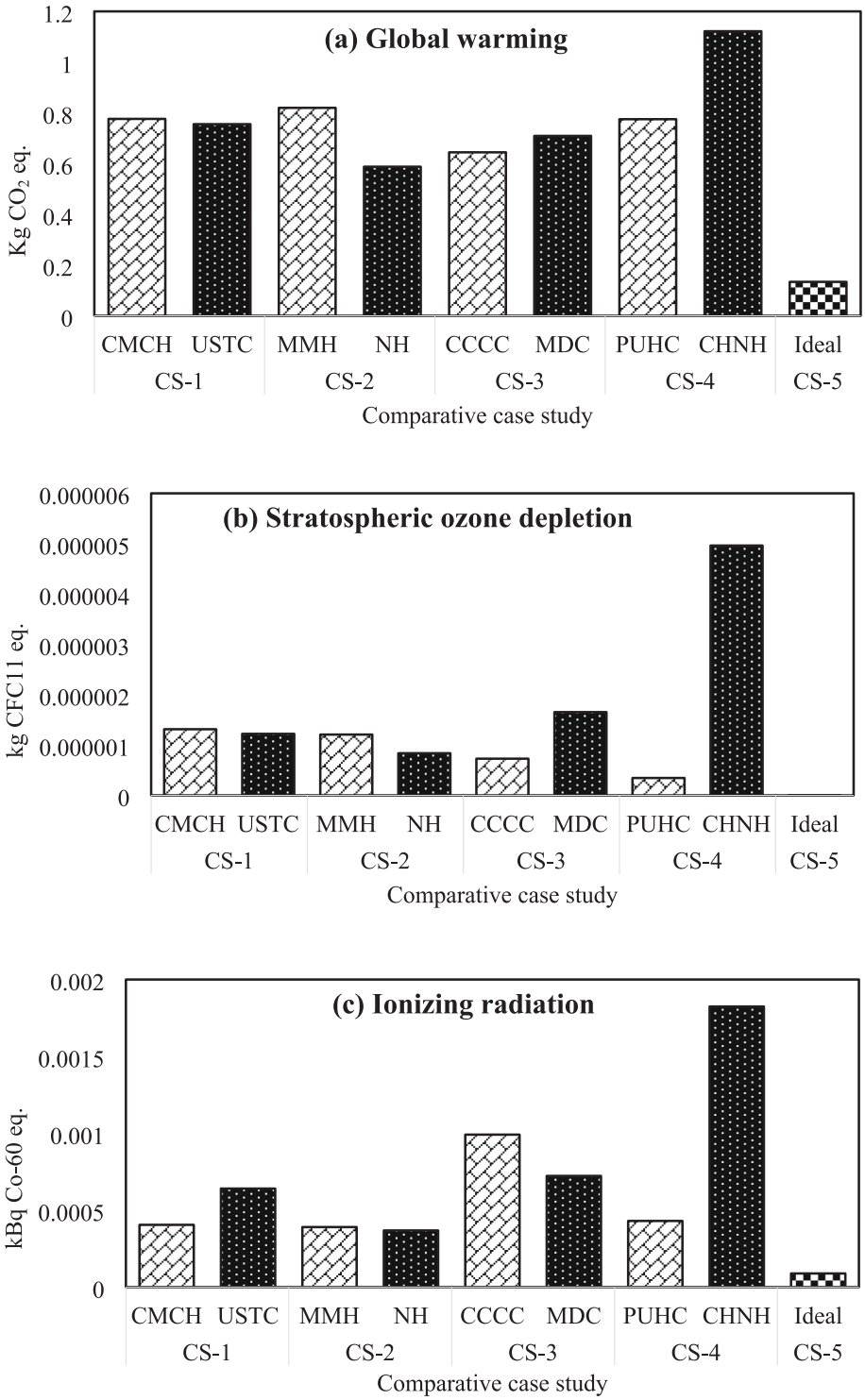



**Fig. 3.** Comparative environmental impact of HCWM system between public and private healthcare facilities.

environmental impacts through LCA (see Table SI-7 (Supporting Information). It showed the major contribution to the studied environmental impacts from HCW, followed by transport (diesel), residues, wood ash and electricity. The main impact through the proposed HCWM system includes resource scarcity, human toxicity, ecotoxicity, eutrophication, acidification and global warming (see Fig. 4).

These impact categories were further categorized into three endpoint impacts to measure damages and losses including human health, ecosystem quality, and resources. The midpoint categories are useful in interpreting the potential contributors to the endpoint category results during the interpretation stage of the LCA analysis. The endpoint environmental damage results are shown in Fig. SI-2 (Supporting Information) and Table SI-8 (Supporting Information). It demonstrated the highest damage in resources, followed by human health and ecosystems. The human health and ecosystem damages are very low, indicating insignificant damages. As large scale HCF, textile waste and wastepaper highly contributed to damages of resources from CMCH while textile waste and waste transport in USTC. Similarly, waste transport, textile

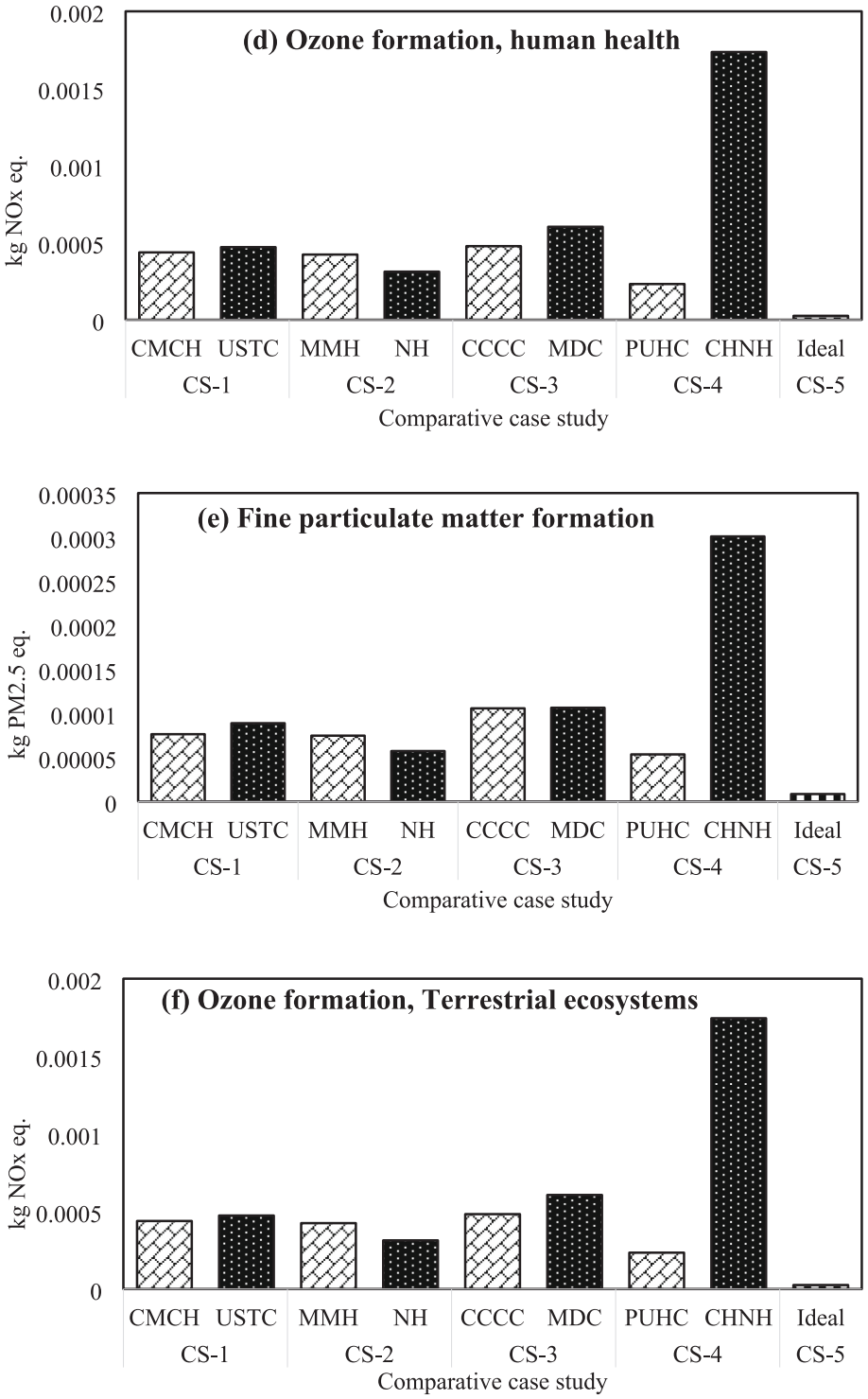



**Fig. 3.** (*continued*).

waste, SPW and wastepaper shared highly to resource damaged from MMH while waste transport, textile waste and SPWs from NH. The transport of HCW along with disposal of waste textile, wastepaper and SPW highly contributed to resource damages from both CCCC and MDC. In addition, waste transport as well textile waste and SPW disposal are shared largely to resource damages from both PUHC and CHNH.

The highest impact was observed in human non-carcinogenic toxicity, followed by global warming, terrestrial ecotoxicity, freshwater ecotoxicity and marine ecotoxicity. The interpretation of the LCA results were conducted by sensitivity analysis. The 20 % increase and 20 % decrease of HCW transport distance demonstrated the increase and decrease of total energy consumption in HCW disposal and treatment. From Fig. 5, it is observed that the change of energy consumption is highly affected in global warming potential. However, the other impact categories (human non-carcinogenic toxicity, terrestrial ecotoxicity, freshwater ecotoxicity and marine ecotoxicity) were comparatively lower but not insignificant.

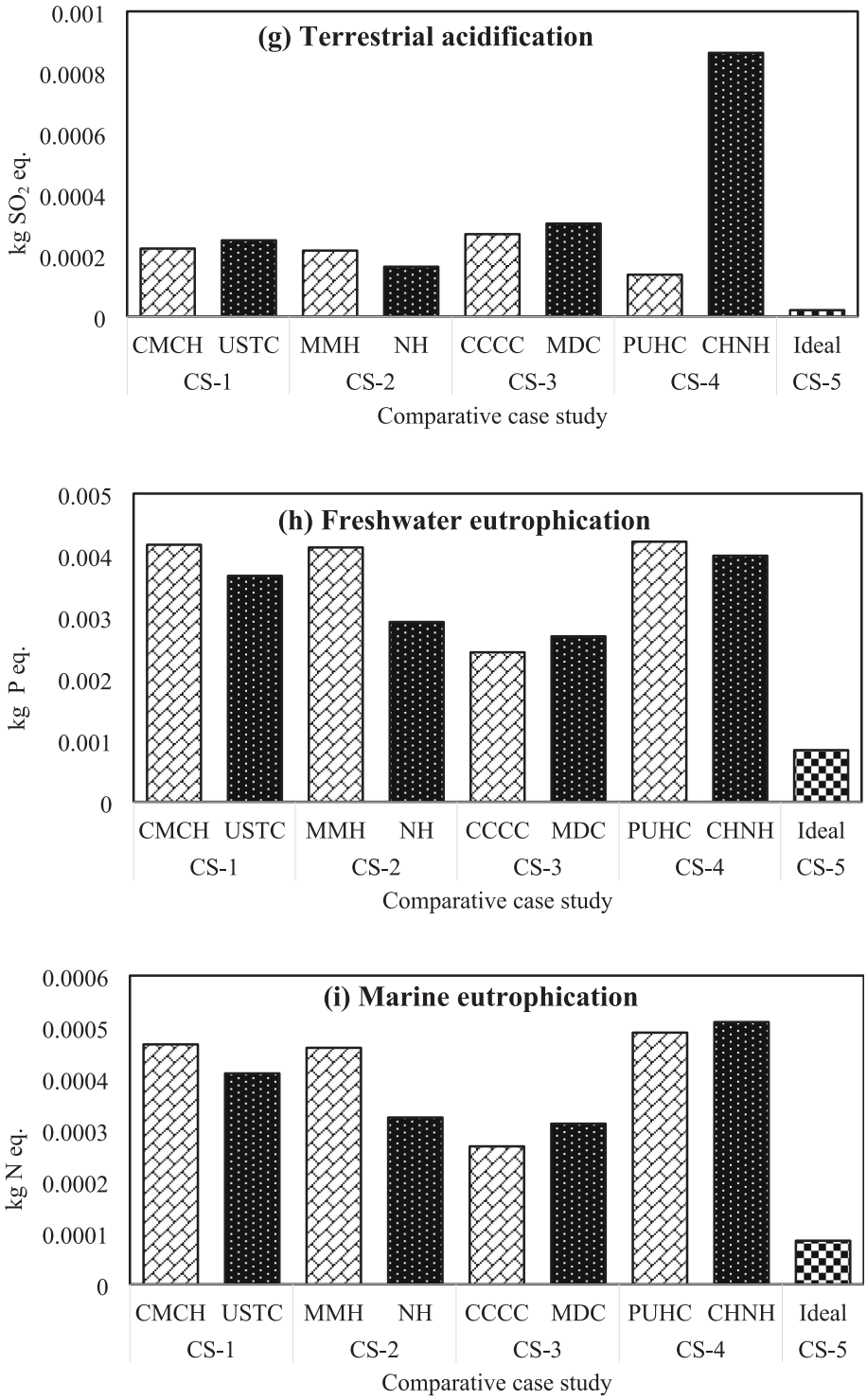
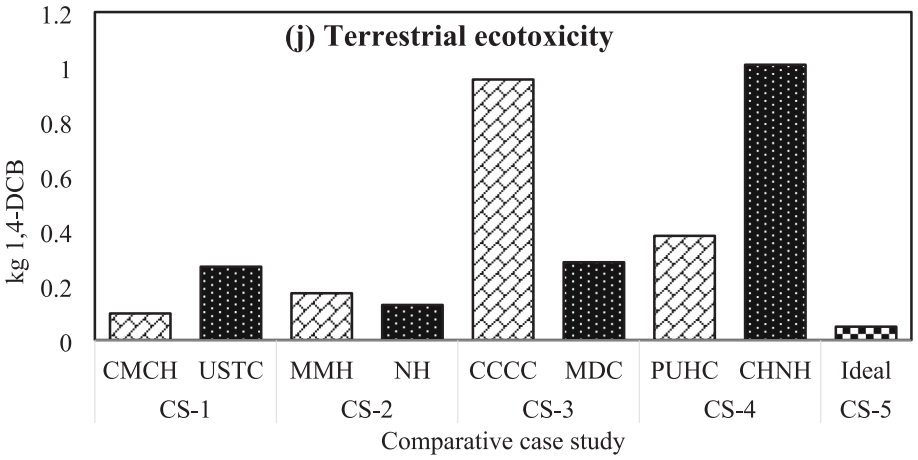


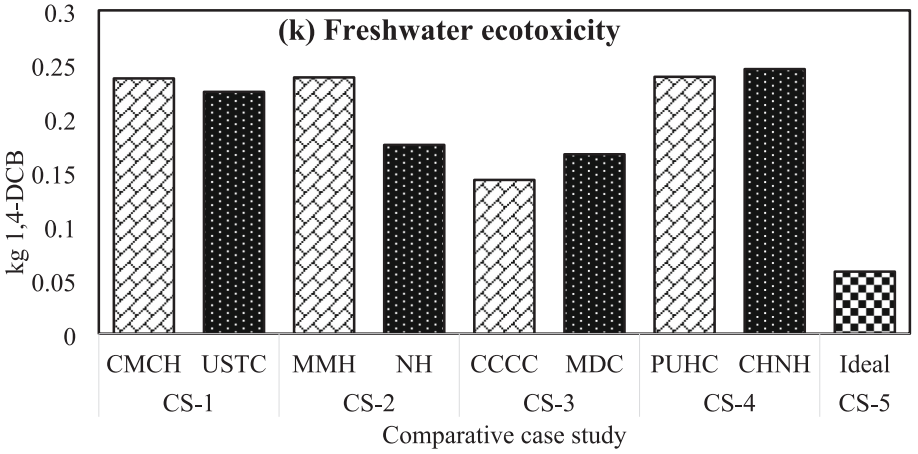

**Fig. 3.** (*continued*).

### 3.2. Microplastics pollution from healthcare waste management

This study demonstrated that the disposal of SPWs largely contributed to global warming, ecotoxicity and carcinogenic toxicity than other studied impact categories that mainly come from the recycling and open burning of SPWs (see Table 3). However, the other impact categories from SPWs that are considered in this LCA study are negligible. Dumping or burial of the daily generated SPWs also contribute to these impacts from the rural HCFs. From Table 3, it is stark clear that GW, TE, Fec, Mec HCT and HNCT are comparatively higher from the MDC that was for the higher percentage of SPW disposal through open burning.

Though the LCA is unable to evaluate the MPs pollution by the existing database; however, the potential MPs pollution from HCWM can be estimated by considering the available relevant literature. Therefore, the plastics HCWM is a potential source of MPs pollution. This investigation identified seven potential sources of MPs pollution including illegal dumping in drains/ lands, scrap shop (crushing), recycling factory, plastics manufacturing factory, open burning/incineration, dumpsite/landfill and sea beach (see Fig. SI-3 (Supporting Information)). At first, MPs are deposited in the wastewater at recycling



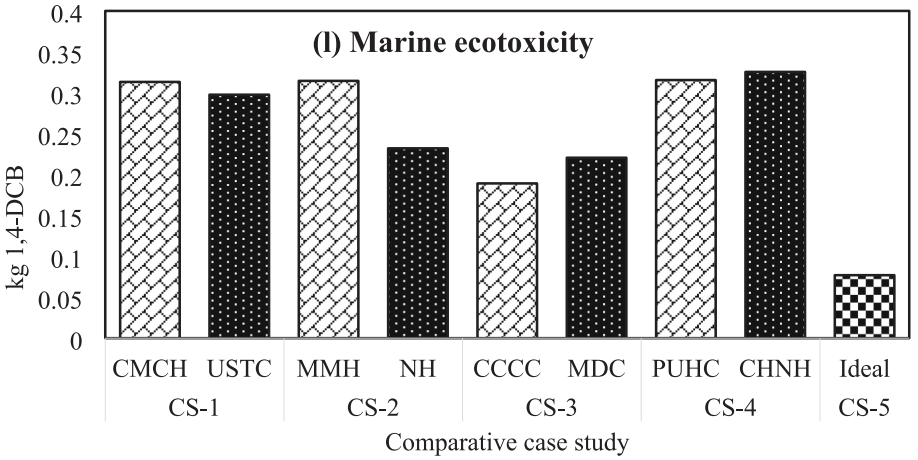




**Fig. 3.** (*continued*).

and manufacturing factories, and then transferred to river water through drainage flow. Similarly, MPs are deposited in the leachate from landfill which subsequently move to the adjacent waterbodies and then river/sea water. The generated MPs are accumulated in bottom ash during open burning and incineration of HCW. Thus, the mishandled plastics HCW contribute to global MPs pollution.

The disposal and treatment of SPW can generate MPs as secondary wastes. The potential MPs emissions from HCW management and disposal are shown in Fig. 6 and Table SI-9 (Supporting Information). The highest MPs generation was observed in private HCFs due to the higher percentage of SWP. As HCW disposal option, the recycling of SPWs through mechanical (to remanufacture polymers) treatment generated the highest quantity of MPs ranging from 1335 to 3106 items/kg waste. Therefore, the mechanical recycling of SPW is one of the major sources of MPs pollution. The generated MPs during the recycling are deposited in the wastewater. Similarly, the crude dumping of SPW in the dumpsite contributed to MPs pollution ranging 291–677 items/kg waste through leachate. In addition, the open burning or incineration of SPWs also released MPs through bottom ash and air emissions that ranged from 280 to 651 items/kg waste. The ideal HCWM system demonstrated

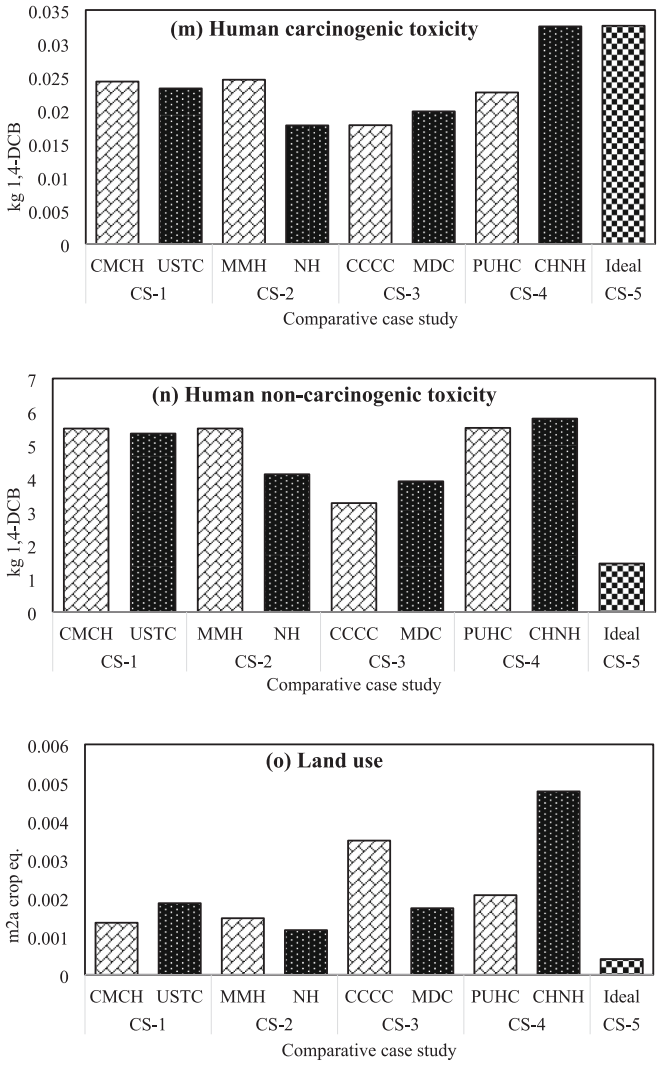



**Fig. 3.** (*continued*).

the insignificant amount of MPs emissions from sanitary landfill and incineration; however, a significant quantity (2191 items/kg waste) of MPs emission from recycling.

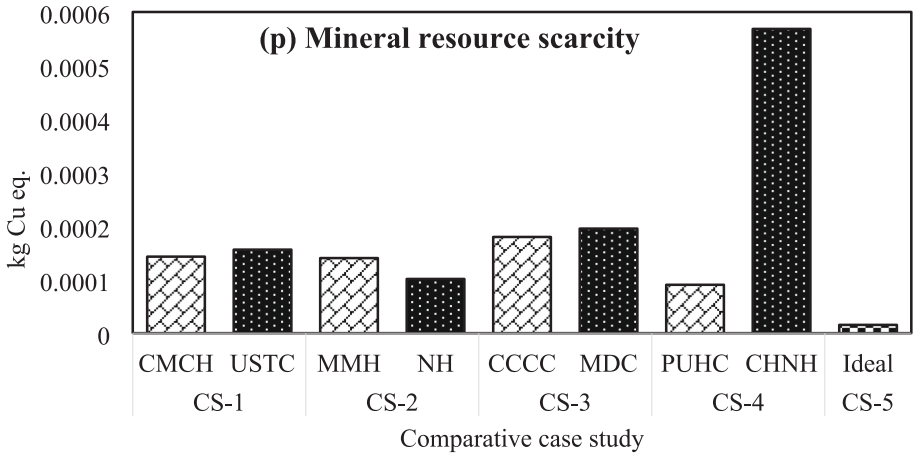
### 3.3. Heavy metals pollution from healthcare waste management

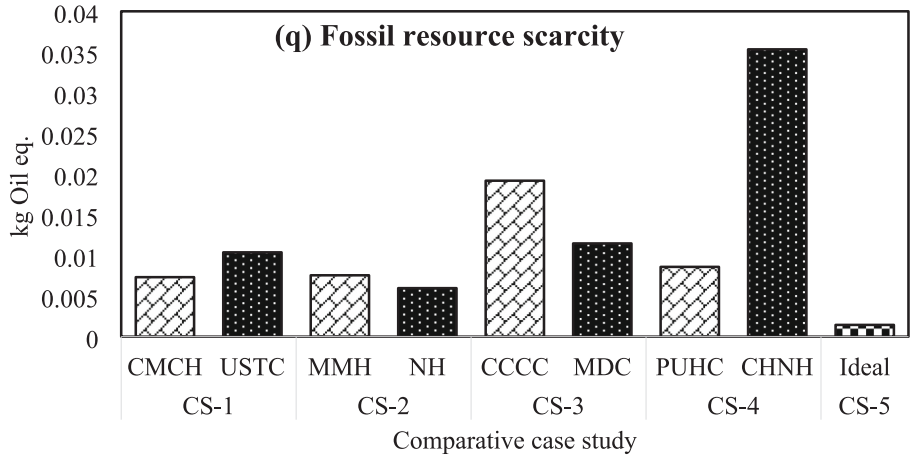
HCW is typically a complex mixture of different types of solid wastes. Most of the inorganic HCWs contain toxic HMs. Such HCW include thermometers, chemicals, plastics, PPE, syringes, needles, rubber, scalpels, metal cans, glass, battery and metallic tools. This research found several HMs in HCW such as Sb, As, Cd, Cr, Cu, Fe, Pb, Mn, Hg, Ni, Se, Ag, and Zn. Amidst, toxic HMs like As, Cd, Pb and Hg, are mostly present in plastic, syringes, rubber, adhesive plaster and battery. These HMs are transferred to the bottom ash when HCW are openly burned or incinerated. Incineration or open burning of HCW with high metallic content (e.g., Pb, Hg and Cd) can lead to the spread of toxic metals in the environment. Therefore, the treatment of HCW either by incinerating or open burning is a potential source of HMs pollution.

The concentrations of HMs are very high in HCW due to containing a major fraction of plastics and metallic wastes. If such waste is illegally dumped, then these HMs are deposited in the leachate and adjacent waterbodies. The leachate flow transfers deposited HMs to surrounding surface water and soil that subsequently reach to ground water and rivers. The recycling of HCW generates HMs through wastewater discharge. Thus, treatment and disposal of HCW shared to HMs pollution (see Fig. SI-4 (Supporting Information)). The results demonstrated that water is at risk of highly HMs pollution and then air and soil (see Table SI-10 (Supporting Information)). Finally, these HMs enter the food chain through the fish and crops/vegetables grown in the polluted water and soil. Here, comparatively high levels of the HMs release to the air, water and soil were from CHNH because it is a rural private clinic which disposed of the daily generated HCW illegally. Those who collect recyclable waste from CHNH recycle waste without any pretreatment. In addition, CMCH, USTC, MDC and PUHC demonstrated comparatively high levels of Fe & Hg, As & Mn, Cr, and Cd, Pb & Sn, respectively release to water from HCW disposal and treatment.

### 3.4. Discussion

LCA is not very old in developing countries. Lately, the researchers are trying to implement it in HCWM of developing countries, but the key problem in conducting LCA study is the absence of a database based on the actual scenarios of such countries (Alam and Mosharraf, 2019; Dey and Ray, 2020, 2021; Snigdha et al., 2023). This study compared existing HCWM systems between public and private HCFs through LCA. It demonstrated that private HCFs contributed higher than public ones in most of the studied impact categories except CMCH (prime HCF in Chattogram district). The inadequate HCWM system in private HCFs is highly responsible for surrounding water, soil and air pollution. The waste handlers are mainly affected by inadequate HCWM in low-income



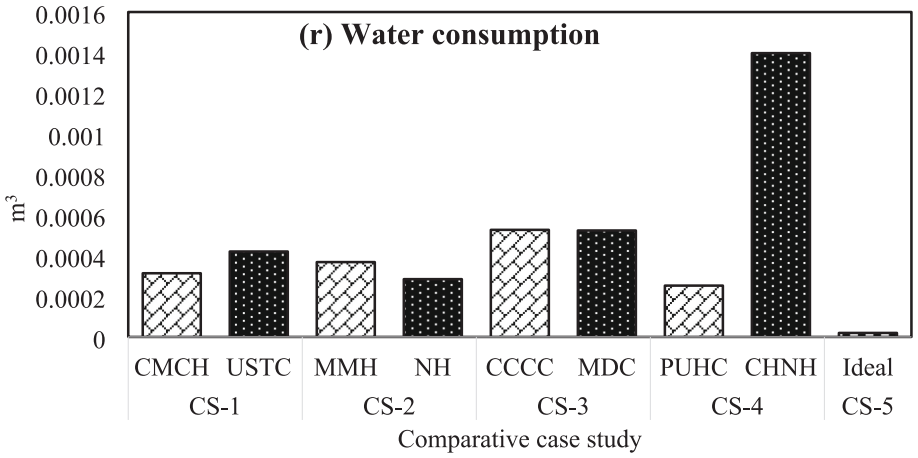




**Fig. 3.** (*continued*).

communities. However, such impact is not so severe in public HCFs due to source segregation of wastes and wearing PPEs by waste handlers and engaging third parties for disposal of the daily generated HCW. Results demonstrated comparatively higher environmental footprints from private HCFs. Therefore, this study recommends some initiative to minimize environmental footprints including strict waste segregation (food waste, sharp, hazardous and recyclable) at sources by using color coding bins, reusable waste disinfection by steam sterilization, and mobile or small-scale incinerator for quick waste disposal (or take support from third party who provide such services by protecting environment).

Notably, the private HCFs always give priority on economic benefits; therefore, they consider spending budget and time in ecofriendly HCWM as wastage. As a result, they are greatly contributing to global warming, ecotoxicity, eutrophication and human toxicity etc. Because the physical composition of HCW of large-scale public HCF is diversified and contains a comparatively higher percentage of infectious and hazardous waste (e.g., radioactive wastes, chemicals, body fluids and body organs), therefore it demands special treatment as well as additional investment. Otherwise, a similar financial budget is required for the same amount of HCW disposal in both public and private HCFs. HCWM funds come from

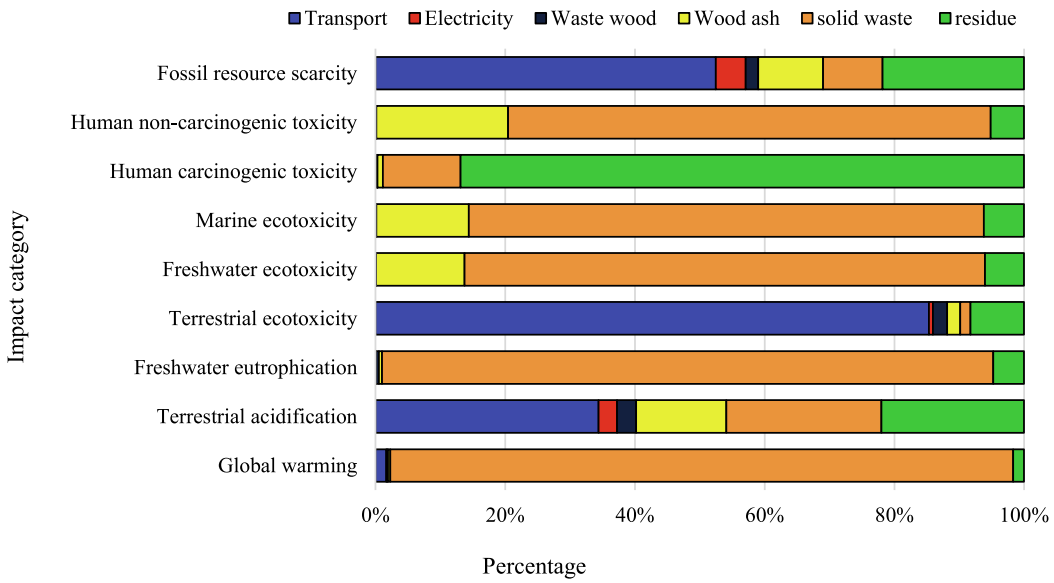



**Fig. 4.** The main environmental impacts from energy and resources consumption in ideal HCWM system.

the government in public HCFs, not from individual persons. However, such a fund needs to be invested by the private HCFs owner himself. Therefore, private HCFs directly discharge their daily generated HCW which demands no cost. In contrast, public HCFs segregate, disinfect, incinerate and dispose of HCW by third party contract which all need a financial budget.

Several researches were conducted to explore the environmental impacts from HCWM, emergency waste disposal and medical test (Ali et al., 2016; Deepak et al., 2022; Ji et al., 2022; Zhao et al., 2021). McPherson et al. (2019) showed carbon footprint from disposable to reusable sharps containers in the United States hospitals. Similarly, several previous studies compared disposable and non-disposable healthcare instruments, surgical masks and PPEs (Kumar et al., 2021; Leiden et al., 2020; Snigdha et al., 2023; Türkmen, 2023). Li et al. (2022) found that treatment of HCW with energy recovery through pyrolysis treatment is comparatively ecofriendly. Yin et al. (2023) and Hong et al. (2018) studied the environmental impacts and economic cost of HCW recycling and treatment. Therefore, LCA based studies are supportive to identify the HCW disposal option with minimum impacts.

HCW is composed of 8 %–20 % plastics based on sources in Bangladesh. These SPWs are inevitably generated from HCFs (Ajaj et al., 2023). The mismanagement of SPW resulted in MPs release that deposited in drains and subsequently waterbodies (Zhang et al., 2021; Zhao and You, 2023). The crude dumping of SPWs along with mechanically or thermally recycling of SPWs shares to MPs pollution (Suzuki et al., 2022) and thus pollutes the aquatic environment. Therefore, SPWs release MPs during its handling, treatment and disposal (Guo et al., 2022). MPs can form volatile organic chemicals via photodegradation and exert significant global warming, ecotoxicity, and air pollution effects. However, recycling of SPWs fast pyrolysis can effectively minimize environmental damage by reducing ozone formation and air pollution by generating fuels and energy (Zhao and You, 2023). Previous studies found the higher MPs pollution impact from the recycling of plastic HCW ranging from 6807 to 28,600 items/ kg plastic waste (Brown et al., 2023; Stapleton et al., 2023). However, the incineration and dumping of HCW showed the comparatively lower MPs pollution. Silva et al. (2021) found up to 291 items/L of the HCW dumping leachate; however, the treatment of leachate can reduce MPs concentration. Similarly, Yang et al. (2021) observed 0.36–102 items/kg waste in bottom ash, while Shen et al. (2021) measured 23, 171, and 86 items/kg in fly ash, bottom ash and soil.

HCW is one of the main sources of global HMs pollution. HMs are particularly released from HCWM during crude dumping, illegal disposal and landfill along with open burning and incineration ash. The dumped HCW make chemical reactions in the presence of sunlight and water, and then release the contained HMs. These HMs are distributed to surrounding water and soil through leachate flow. This study demonstrated that HMs are released to air, water and soil from HCWM systems following the order as water > air > soil. Here, 5.89 μg Cd and 6.49 μg Co emission to air were found as the highest in City corporation clinics (CCCC) while other studied HMs were the highest in rural private clinic (CHNH). In soil emissions, only 0.81 μg Co was the highest from City corporation clinics (CCCC) while other studied HMs were the highest in private rural clinics (CHNH). Similarly, 14480 mg Fe and 0.71 mg Hg from CMCH, 0.8 mg As and 858.1 mg Mn from USTC, 0.03 mg Cr from MDC, 7.82 mg Cd and 259.6 mg Pb from PUHC, and other studied HMs (except Sb and Cr(VI)) from CHNH in water emissions. In addition, 7.58 mg Sb and 4.32 mg Cr(VI) from the ideal HCWM system. Several studies demonstrated the high level of HMs content in HCW dumped soil and water as well released leachate (Dey and Ray, 2023; Janik-Karpinska et al., 2023). HMs from HCW are transferred to ash during the open burning or incineration. Therefore, previous studies identified excess levels of toxic HMs in burnt or incinerated ash. Regretfully, such HMs containing ash are disposed of in landfill without pretreatment in developing countries like Bangladesh (Adama et al., 2016; Dey and Ray, 2021; Zhao et al., 2009). However, such HMs containing ash are disposed of in sanitary landfill after treatment or treated for reusing in construction or land development such as solidification/ stabilization in developed countries.

This LCA study compared four categories of HCFs. The composition of HCW generated from these HCFs varied and therefore their management and disposal system differed too. The findings of this comparative study in terms of environmental and health impacts helpful to adopt a generalized HCWM system with minimum impacts. Besides, this study demonstrated the higher MPs pollution from HCWM, which is a global emerging environmental issue. According to this study, about 100 % plastic waste is recycled from private HCFs. Plastics HCW from private HCFs are directly discharged or illegally collected by waste pickers (scavengers) and then sold to recycling shops. Therefore, MPs emission is also high from private HCFs. At first, plastics wastes are mechanically ground when MPs are generated. Then ground plastics are melted to form plastic beads or flakes. Finally, MPs are generated during producing recycled products. Thus, generated MPs are ultimately deposited to wastewater recycling plants. Similarly, the incomplete

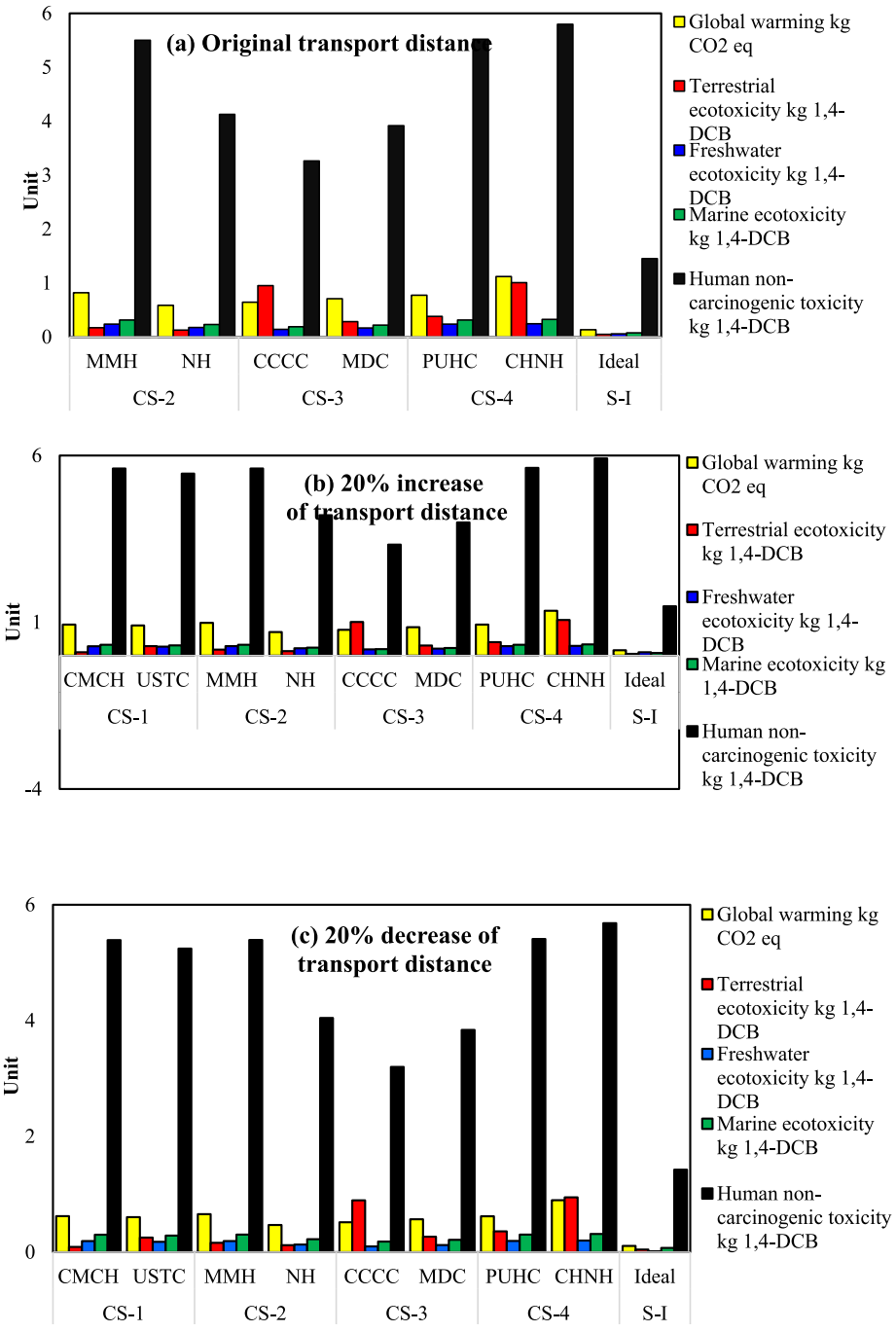



**Fig. 5.** Sensitivity analysis based on the change of HCW transport distance to disposal site.

**Table 3**
Major impact categories from plastic waste disposal through life cycle assessment.

| Impact category | Unit | CS-1 | | CS-2 | | CS-3 | | CS-4 | |
|---|---|---|---|---|---|---|---|---|---|
| | | CMCH | USTC | MMH | NH | CCCC | MDC | PUHC | CHNH |
| GW | kg $CO_2$ eq | 0.02 | 0.07 | 0.08 | 0.00 | 0.00 | **0.17** | 0.08 | 0.12 |
| TE | kg 1,4-DCB | 0.01 | 0.04 | 0.04 | 0.01 | 0.01 | **0.09** | 0.05 | 0.06 |
| Fec | kg 1,4-DCB | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | **0.01** | 0.01 | 0.01 |
| Mec | kg 1,4-DCB | 0.00 | 0.01 | 0.01 | 0.00 | 0.00 | **0.01** | 0.01 | 0.01 |
| HCT | kg 1,4-DCB | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | **0.00** | 0.00 | 0.00 |
| HNT | kg 1,4-DCB | 0.02 | 0.05 | 0.06 | 0.02 | 0.02 | **0.12** | 0.06 | 0.09 |

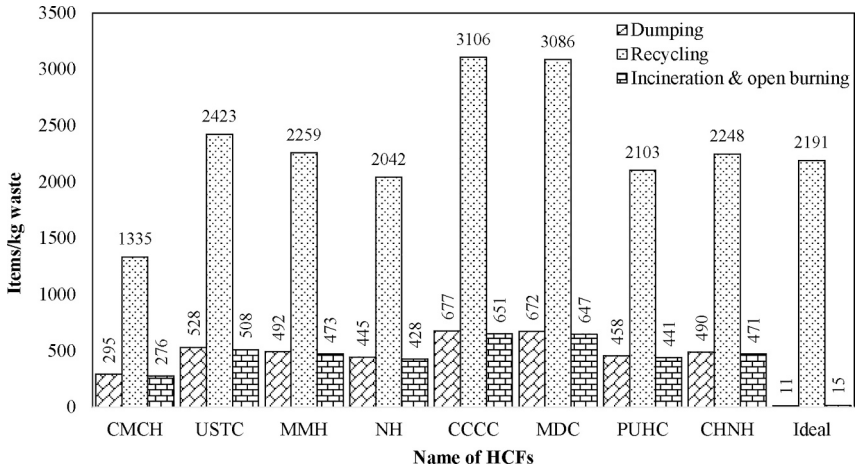



**Fig. 6.** Contribution of microplastics by different HCW disposal and treatment options in total microplastics quantity. Note: MPs emissions were counted by comparing same amount of plastics wastes as input for LCA software and the MPs emission from same amount of plastics wastes disposal as the output ( Adopted from Brown et al., 2023; Salami and Popoola, 2000; Shen et al., 2021; Silva et al., 2021; Stapleton et al., 2023; Yang et al., 2021).

burning of plastics HCW generates MPs which are deposited in bottom ash and lighter MPs are released to air. Furthermore, small plastic pieces are generated by degrading larger plastics in dumping sites due to the influence of several factors including UV radiation, fluctuating temperatures (60–90 °C) and pH (4.5–9.0), deep-seated fires, physical stress, and compaction, along with limited microbial activity. Thus, the management of plastics waste separately or mixing with other wastes contribute to MPs emissions. These findings show the potential way for reducing the MPs emissions from the HCWM system.

Similarly, this study showed the HMs pollution from the HCWM system. The most common detrimental HMs that come the HCWM system include As, Cd, Cr, Cu, Pb, Hg, Ni. Here, Pb, Cd and Hg comes from high-metals-contents wastes incinerated or open burned ashes. Besides, Fe, Mn and Cu are also present in incinerated bottom ashes. Volatile metals, such as Hg, Pb, As and Cd may pollute air during incineration/open burning of HCW. These HMs come from a specific type of HCW. For example, Cd, Hg, Pb and Zn from batteries, Ni and Cr from stainless steel needles, and Cu, Pb, Cr and Cd from plastics. Slightly Pb and Cd may come from glasses. In addition, Pb, Ni, Cr, Cu, Zn and Cd may come from food wastes disposal through leachate. The results showed that HMs are highly released from open burning or incineration of HCW. Generally, there are three methods for controlling HMs emissions from HCW incineration or burning: (i) limiting the HM contents of the waste feed via source control, (ii) designing and operating the incineration process to minimize HM vaporization, (iii) designing and operating the primary incineration chamber to minimize fly-ash carryover. Besides, controlled-air incineration is the most widely used HCW incinerator technology. The mechanical recycling of recyclable HCW after proper treatment can minimize HMs emissions to the environment. However, wastewater recycling plants can contain HMs. Similarly, leachate from organic HCW dumping can also release HMs to the environment. However, HMs in wastewater and leachate can be removed by chemical precipitation.

The potential primary MPs sources from plastics HCWM include mechanical recycling, incineration/open burning and illegal dumping. From recycling and incineration of plastic wastes, humans are exposed to MPs through air. However, humans are highly exposed to MPs through wastewater (recycling plant), incineration/burnt ashes, and leachate from dumping sites. Exposure to MPs in humans potentially leads to serious health issues, including various cancers, respiratory disorders, and inflammatory bowel disease. Understanding molecular mechanisms underlying the interaction of MPs with human cells may help extrapolate the risks to humans. There are several potential sources of HMs emissions from HCWM including dumping, open burning/incineration and recycling. From these sources, HMs are released to water and soil. Furthermore, some HMs such as Pb and Hg are also released to air. From the environment HMs enter the human body through drinking water and taking foods grown in such contaminated soil/water, and directly inhaled from air. HMs can cause a variety of health problems in humans, including organ damage, cancer, and neurological disorders. Bioaccumulation of these HMs leads to a diversity of toxic effects on a variety of body tissues and organs. HMs disrupt cellular events including growth, proliferation, differentiation, damage-repairing processes, and apoptosis.

Considering the higher impacts from private HCFs, it is advisable for them to adopt strategies with minimum environmental footprints, HMs and MPs emissions from HCWM systems. In private HCFs, the adoption of appropriate source segregation of HCW, stream disinfection of infectious and hazardous waste, controlled incineration and recycling of HCW can minimize environmental emissions from HCWM systems. The suggested strategies cannot be easily executed in HCFs, specifically in private HCFs. It will face several challenges such as sufficient funds, proper knowledge, skilled manpower, and advanced technologies. Therefore, it is required to provide free training to associated workers and staff of private HCFs by government or other concerned authorities. Government should give incentives to HCFs for adopting HCWM technologies in an eco-friendly way. The government formulated latest related regulations which also compel HCFs to adopt such strategies for safe HCWM. However, the poor implementation of such regulations discourages HCFs to adopt these strategies, and therefore these regulations should be implemented strictly. These strategies can be effectively replicated in similar other HCFs and urban areas throughout the country and other developing countries.

Therefore, these findings show the potential environmental impacts, and level of MPs and HMs emissions which can facilitate in adopting the required controlling strategies. However, the available LCA software cannot measure MPs pollution due to the absence of a relevant database. Therefore, this study used the data from some relevant studies to calculate MPs emissions from HCWM.

## 4. Conclusion and future research direction

HCWM is a globally environmental concern. The developing countries like Bangladesh are at high risk due to possessing a major fraction of hazardous waste. Regretfully, these wastes are illegally disposed of with MSW in such countries that are liable for environmental pollution. Recently, LCA is considered as the best evaluation tool for environmental burdens associated with HCWM. This study reveals that the current HCWM system differed based on sources and category of HCFs.

Private HCFs highly contributed to most of the studied impact categories (e.g., 1.12 kg $CO_2$ eq. global warming, 1.01 kg 1,4-DCB terrestrial ecotoxicity and 5.8 kg 1,4-DCB human non-carcinogenic toxicity) than public ones. It might be due to the uncontrolled and illegal disposal of all HCW by private HCFs, especially in rural areas where HCWM facilities are totally absent. Here, illegal recycling and open burning/ incineration are mainly responsible for human toxicity and global warming, while crude dumping and landfill are liable for ecotoxicity and eutrophication. The existing HCWM system can release HMs through illegal dumping, recycling plant wastewater and incineration ash. The water environment is more risk at HMs pollution than air and soil environments. Furthermore, this study demonstrated the higher quantity of MPs emission from the recycling of HCW that are accumulated in wastewater. However, the factors responsible for MPs formation and their associated fate in the environment is still unexplored. Though it is possible to measure the impacts of SPWs through LCA but there is no specific way to measure MPs emission from different stages of HCWM. So, it is strongly suggested to develop LCA databases using actual scenarios of developing countries.

This study mainly used MPs data from literature. Plastic wastes are almost 100 % recycled in low-income countries like Bangladesh. Though HCW recycling is prohibited by WHO or other related international organizations. Currently, there is no LCA database for MPs emission from HCW disposal. Therefore, this study proposes some future research directions to measure MPs emissions from plastic HCW disposal and subsequently, the emissions data will be used for LCA database formulation, specifically for low-income communities. For MPs emission from plastic waste incineration, in a controlled way plastic waste will be incinerated to collect air samples and bottom ash. Then both samples will be analyzed in the laboratory for counting MPs contents in samples. Similarly, both air and wastewater samples will be collected from a recycling plant to measure MPs content which subsequently will be analyzed in the laboratory. In addition, mixed HCW including plastic waste will be dumped in sanitary landfill and dumping site, then leachate will be collected from both disposal points with time intervals 7 days, 30 days, 90 days, 180 days and 365 days for MPs emissions counting both in dry and wet seasons. Finally, these experimental data of MPs from disposal of plastics HCW will be applied for related LCA database development. It is recommended that future studies should include a sensitivity analysis to assess the potential impact of excluding wastewater, gaseous emissions, and upstream processes. This analysis could provide insights into how these exclusions might affect the overall environmental impact assessment. For example, hypothetical scenarios could be developed to estimate the potential contributions of these excluded stages based on available literature or expert judgment. Similarly, future studies should consider a more comprehensive LCA that includes upstream processes. This would provide a more holistic view of the environmental impacts associated with HCWM, from product manufacturing to final disposal.

## CRediT authorship contribution statement

**Xiaojun Zheng:** Writing – original draft, Conceptualization. **Shan Zhong:** Writing – review & editing, Supervision. **Ohidul Alam:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Syed Jawad Hossen:** Methodology, Formal analysis. **Daolin Du:** Writing – review & editing.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Daolin Du reports financial support was provided by the Carbon Peak and Carbon Neutrality Technology Innovation Foundation of Jiangsu Province. Shan Zhong reports financial support was provided by the Scientific Research Foundation for Senior Talent of Jiangsu University, China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.wasman.2025.114932.

## Data availability

Data will be made available on request.

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